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Electronic Properties of Materials

Fourth Edition



CHAPTER 3

The Schrödinger Equation

We shall now make use of the conceptual ideas which we introduced in the previous chapter, i.e., we shall cast, in mathematical form, the description of an electron as a wave, as suggested by Schrödinger in 1926. All “derivations” of the Schrödinger equation start in one way or another from certain assumptions, which cause the uninitiated reader to ask the legitimate question, “Why just in this way?” The answer to this question can naturally be given, but these explanations are relatively involved. In addition, the “derivations” of the Schrödinger equation do not further our understanding of quantum mechanics. It is, therefore, not intended to “derive” here the Schrödinger equation. We consider this relation as a fundamental equation for the description of wave properties of electrons, just as the Newton equations describe the matter properties of large particles.

3.1. The Time-Independent Schrödinger Equation

The *time-independent* Schrödinger equation will always be applied when the properties of atomic systems have to be calculated in stationary conditions, i.e., when the property of the surroundings of the electron does not change with time. This is the case for most of the applications which will be discussed in this text. Thus, we introduce, at first, this simpler form of the Schrödinger equation in which the potential energy (or potential barrier), V , depends only on the location (and not, in addition, on the time). Therefore, the time-independent Schrödinger equation is an equation of a *vibration*. It has the following form:

$$\boxed{\nabla^2 \psi + \frac{2m}{\hbar^2} (E - V) \psi = 0,} \quad (3.1)$$

where

$$\nabla^2 \psi = \frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2}, \quad (3.2)$$

and m is the (rest) mass of the electron,² and

$$E = E_{\text{kin}} + V \quad (3.3)$$

is the total energy of the system. E provides values for allowed energies once ψ and V are given, as we shall see later on.

In (3.1) we wrote for the wave function a lowercase ψ , which we will use from now on when we want to state explicitly that the wave function is only space dependent. Thus, we split from Ψ a time-dependent part:

$$\Psi(x, y, z, t) = \psi(x, y, z) \cdot e^{i\omega t}. \quad (3.4)$$

*3.2. The Time-Dependent Schrödinger Equation

The *time-dependent* Schrödinger equation is a *wave* equation, because it contains derivatives of Ψ with respect to space *and* time (see below, (3.8)). One obtains this equation from (3.1) by eliminating the total energy,

$$E = \nu h = \omega \hbar, \quad (2.1)$$

where ω is obtained by differentiating (3.4) with respect to time:

$$\frac{\partial \Psi}{\partial t} = \psi i \omega e^{i\omega t} = \Psi i \omega. \quad (3.5)$$

This yields

$$\omega = -\frac{i}{\Psi} \frac{\partial \Psi}{\partial t}. \quad (3.6)$$

Combining (2.1) with (3.6) provides

$$E = -\frac{\hbar i}{\Psi} \frac{\partial \Psi}{\partial t}. \quad (3.7)$$

²In most cases we shall denote the rest mass by m instead of m_0 .

Finally, combining (3.1) with (3.7) yields

$$\boxed{\nabla^2 \Psi - \frac{2mV}{\hbar^2} \Psi - \frac{2mi}{\hbar} \frac{\partial \Psi}{\partial t} = 0.} \quad (3.8)$$

It should be noted here that quantum mechanical equations can be obtained from classical equations by applying differential operators to the wave function Ψ (Hamiltonian operators). They are

$$E \hat{=} -\hbar i \frac{\partial}{\partial t} \quad (3.9)$$

and

$$\mathbf{p} \hat{=} -\hbar i \nabla. \quad (3.10)$$

When these operators are applied to

$$E_{\text{total}} = E_{\text{kin}} + E_{\text{pot}} = \frac{p^2}{2m} + V \quad (3.11)$$

we obtain

$$-\hbar i \frac{\partial \Psi}{\partial t} = \frac{\hbar^2 \nabla^2}{2m} \Psi + V \Psi, \quad (3.12)$$

which yields, after rearranging, the time-dependent Schrödinger equation (3.8).

*3.3. Special Properties of Vibrational Problems

The solution to an equation for a vibration is determined, except for certain constants. These constants are calculated by using boundary or starting conditions

$$(\text{e.g., } \psi = 0 \text{ at } x = 0). \quad (3.13)$$

As we will see in Section 4.2, only certain vibrational forms are possible when boundary conditions are imposed. This is similar to the vibrational forms of a vibrating string, where the fixed ends cannot undergo vibrations. Vibrational problems that are determined by boundary conditions are called **boundary or eigenvalue problems**. It is a peculiarity of vibrational problems with boundary conditions that not all frequency values are possible and, therefore, because of

$$E = v h, \quad (3.14)$$

not all values for the energy are allowed (see next chapter). One calls the allowed values eigenvalues. The functions ψ , which belong to the eigenvalues

and which are a solution of the vibration equation and, in addition, satisfy the boundary conditions, are called **eigenfunctions** of the differential equation.

In Section 2 we related the product $\psi\psi^*$ (which is called the “**norm**”) to the probability of finding a particle at a given location. The probability of finding a particle somewhere in space is *one*, or

$$\int \psi\psi^* d\tau = \int |\psi|^2 d\tau = 1. \quad (3.15)$$

Equation (3.15) is called the normalized eigenfunction.

Problems

1. Write a mathematical expression for a vibration (vibrating string, for example) and for a wave. (See Appendix 1.) Familiarize yourself with the way these differential equations are solved. What is a “trial solution?” What is a boundary condition?
2. Define the terms “vibration” and “wave”.
3. What is the difference between a damped and an undamped vibration? Write the appropriate equations.
4. What is the complex conjugate function of:
 - (a) $\hat{x} = a + bi$; and
 - (b) $\Psi = 2Ai \sin \alpha x$.

CHAPTER 4

Solution of the Schrödinger Equation for Four Specific Problems

4.1. Free Electrons

At first we solve the Schrödinger equation for a simple but, nevertheless, very important case. We consider electrons which propagate freely, i.e., in a potential-free space in the positive x -direction. In other words, it is assumed that no “wall,” i.e., no potential barrier (V), restricts the propagation of the electron wave. The potential energy V is then zero and the Schrödinger equation (3.1) assumes the following form:

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}E\psi = 0. \quad (4.1)$$

This is a differential equation for an undamped vibration³ with spatial periodicity whose solution is known to be³

$$\psi(x) = Ae^{i\alpha x}, \quad (4.2)$$

where

$$\alpha = \sqrt{\frac{2m}{\hbar^2}E}. \quad (4.3)$$

(For our special case we do not write the second term in (A.5)³,

$$u = Ae^{i\alpha x} + Be^{-i\alpha x}, \quad (4.4)$$

because we stipulated above that the electron wave³

$$\Psi(x) = Ae^{i\alpha x} \cdot e^{i\omega t} \quad (4.5)$$

³See Appendix 1.



Figure 4.1. Energy continuum of a free electron (compare with Fig. 4.3).

propagates only in the positive x -direction and not, in addition, in the negative x -direction.)

From (4.3), it follows that

$$E = \frac{\hbar^2}{2m} \alpha^2. \quad (4.6)$$

Since no boundary condition had to be considered for the calculation of the free-flying electron, all values of the energy are “allowed,” i.e., one obtains an **energy continuum** (Fig. 4.1). This statement seems to be trivial at this point. The difference to the bound electron case will become, however, evident in the next section.

Before we move ahead, let us combine equations (4.3), (2.3), and (1.4), i.e.,

$$\alpha = \sqrt{\frac{2mE}{\hbar^2}} = \frac{p}{\hbar} = \frac{2\pi}{\lambda} = k, \quad (4.7)$$

which yields

$$E = \frac{\hbar^2}{2m} k^2. \quad (4.8)$$

The term $2\pi/\lambda$ was defined in (2.5) to be the wave number, k . Thus, α is here identical with k . We see from (4.7) that the quantity k is proportional to the momentum p and, because of $\mathbf{p} = m\mathbf{v}$, also proportional to the velocity of the electrons. Since both momentum and velocity are vectors, it follows that k is a vector, too. Therefore, we actually should write k as a vector which has the components k_x , k_y , and k_z :

$$|\mathbf{k}| = \frac{2\pi}{\lambda}. \quad (4.9)$$

Since $\mathbf{k}$ is inversely proportional to the wavelength, λ , it is also called the “wave vector.” We shall use the wave vector in the following sections frequently. The $\mathbf{k}$ -vector describes the wave properties of an electron, just as one describes in classical mechanics the particle property of an electron with the momentum. As mentioned above, $\mathbf{k}$ and $\mathbf{p}$ are mutually proportional, as one can see from (4.7). The proportionality factor is $1/\hbar$.

4.2. Electron in a Potential Well (Bound Electron)

We now consider an electron that is bound to its atomic nucleus. For simplicity, we assume that the electron can move freely between two infinitely high potential barriers (Fig. 4.2). The potential barriers do not allow the electron to escape from this potential well, which means that $\psi = 0$ for $x \leq 0$ and $x \geq a$. We first treat the one-dimensional case just as in Section 4.1, i.e., we assume that the electron propagates only along the x -axis. However, because the electron is reflected on the walls of the well, it can now propagate in the positive, as well as in the negative, x -direction. In this respect, the present problem is different from the preceding one. The potential energy inside the well is zero, as before, so that the Schrödinger equation for an electron in this region can be written, as before,

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}E\psi = 0. \quad (4.10)$$

Because of the two propagation directions of the electron, the solution of (4.10) is

$$\psi = Ae^{i\alpha x} + Be^{-i\alpha x} \quad (4.11)$$

(see Appendix 1), where

$$\alpha = \sqrt{\frac{2m}{\hbar^2}E}. \quad (4.12)$$

We now determine the constants A and B by means of boundary conditions. We just mentioned that at $x \leq 0$ and $x \geq a$ the ψ function is zero. This boundary condition is similar to that known for a vibrating string, which does not vibrate at the two points where it is clamped down. (See also Fig. 4.4(a).) Thus, for $x = 0$ we stipulate $\psi = 0$. Then we obtain from (4.11)

$$B = -A. \quad (4.13)$$

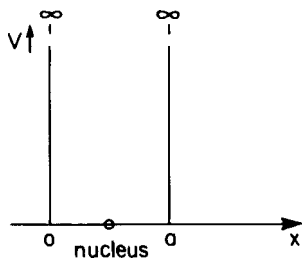


Figure 4.2. One-dimensional potential well. The walls consist of infinitely high potential barriers.

Similarly, we stipulate $\psi = 0$ for $x = a$. Using this boundary condition and (4.13), equation (4.11) becomes

$$0 = Ae^{ixa} + Be^{-ixa} = A(e^{ixa} - e^{-ixa}). \quad (4.14)$$

With the Euler equation

$$\sin \rho = \frac{1}{2i}(e^{i\rho} - e^{-i\rho}) \quad (4.15)$$

(see Appendix 2), we rewrite Equation (4.14)

$$A[e^{ixa} - e^{-ixa}] = 2Ai \cdot \sin \alpha a = 0. \quad (4.16)$$

Equation (4.16) is only valid if $\sin \alpha a = 0$, i.e., if

$$\alpha a = n\pi, \quad n = 0, 1, 2, 3, \dots \quad (4.17)$$

(because 2, A , and i cannot be zero).

Substituting the value of α from (4.12) into (4.17) provides

$$E_n = \frac{\hbar^2}{2m} \alpha^2 = \frac{\hbar^2 \pi^2}{2ma^2} n^2, \quad n = 1, 2, 3 \dots \quad (4.18)$$

(We exclude $n = 0$, which would yield $\psi = 0$, that is, no electron wave.) We notice immediately a striking difference from the case in Section 4.1. Because of the boundary conditions, only certain solutions of the Schrödinger equation exist, namely those for which n is an integer. In the present case the energy assumes only those values which are determined by (4.18). All other energies are not allowed. The allowed values are called “**energy levels**.” They are shown in Fig. 4.3 for a one-dimensional case. Because of the fact that an electron of an isolated atom can assume only certain energy levels, it follows that the energies which are excited or absorbed also possess only discrete values. The result is called an “**energy quantization**.” The lowest energy that an electron may assume is called the “**zero-point energy**”. It can be calculated from (4.18) for $n = 1$. In other words, the lowest energy of the electron is not that of the bottom of the potential well, but rather a slightly higher value.

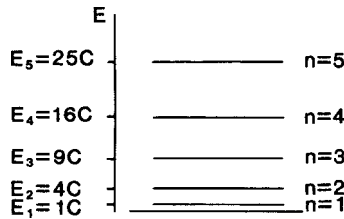


Figure 4.3. Allowed energy values of an electron that is bound to its atomic nucleus. E is the excitation energy in the present case. $C = \hbar^2 \pi^2 / 2ma^2$, see (4.18). (E_1 is the zero-point energy).

We discuss now the wave function, ψ , and the probability $\psi\psi^*$ for finding an electron within the potential well (see Chapter 2). According to (4.11), (4.13), and the Euler equation (4.15), we obtain within the well

$$\psi = 2Ai \cdot \sin \alpha x, \quad (4.19)$$

and the complex conjugate of ψ

$$\psi^* = -2Ai \sin \alpha x. \quad (4.20)$$

The product $\psi\psi^*$ is then

$$\psi\psi^* = 4A^2 \sin^2 \alpha x. \quad (4.21)$$

Equations (4.19) and (4.21) are plotted for various n -values in Fig. 4.4. From Fig. 4.4(a), we see that standing electron waves are created between the walls of the potential well. Note that integer multiples of half a wavelength are equal to the length, a , of the potential well. The present case, in its mathematical treatment, as well as in its result, is analogous to that of a vibrating string.

Of special interest is the behavior of the function $\psi\psi^*$, i.e., the probability of finding the electron at a certain place within the well (Fig. 4.4(b)). In the classical case the electron would travel back and forth between the walls. Its probability function would therefore be equally distributed along the whole length of the well. In wave mechanics the deviation from the classical case is most pronounced for $n = 1$. In this case, $\psi\psi^*$ is largest in the middle of the well and vanishes at the boundaries. For $n = 2$, the probability of finding an electron at the center of the well (and at the boundaries) is zero, whereas the largest $\psi\psi^*$ is found at $\frac{1}{4}a$ and $\frac{3}{4}a$. For successively higher n -values, i.e., for higher energies, the wave mechanical values for $\psi\psi^*$ are eventually approaching the classical value.

In order to deepen the understanding of the behavior of bound electrons, the reader is reminded of the **Rutherford model** (Chapter 2), in which the

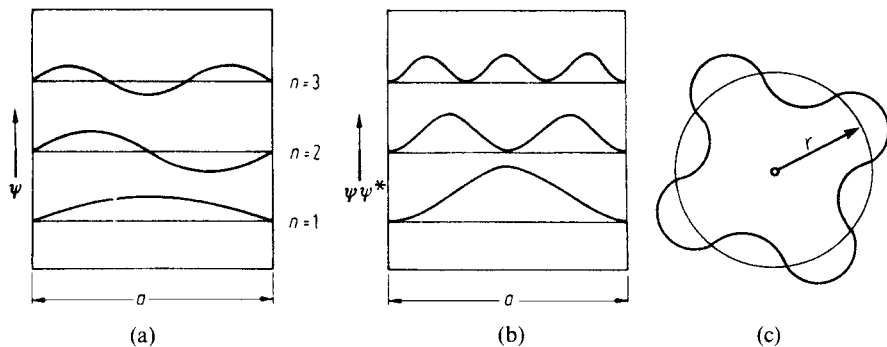


Figure 4.4. (a) ψ function and (b) probability function $\psi\psi^*$ for an electron in a potential well for different n -values. (c) Allowed electron orbit of an atom.

electrons are described to move in distinct orbits about a positively charged nucleus. Similarly, as shown in Fig. 4.4(a), the electron waves associated with an orbiting electron have to be standing waves. If this were not the case, the wave would be out of phase with itself after one orbit. After a large number of orbits, all possible phases would be obtained and the wave would be annihilated by destructive interference. This can only be avoided if a radius is chosen so that the wave joins on itself (Fig. 4.4(c)). In this case the circumference, $2\pi r$, of the orbit is an integer multiple, n , of the wavelength, λ , or

$$2\pi r = n\lambda, \quad (4.22)$$

which yields

$$r = \frac{\lambda}{2\pi} n. \quad (4.23)$$

This means that only certain distinct orbits are allowed, which brings us back to the allowed energy levels which we discussed above. Actually, this model was proposed in 1913 by Niels Bohr.

*For the above discussions, we did not need to evaluate the constant 'A'. Those readers who are interested in this detail may simply rewrite (4.21) in conjunction with (3.15):

$$\int_0^a \psi \psi^* dx = 4A^2 \int_0^a \sin^2(\alpha x) dx = \frac{4A^2}{\alpha} \left[-\frac{1}{2} \sin \alpha x \cos \alpha x + \frac{\alpha x}{2} \right]_0^a = 1. \quad (4.24)$$

Inserting the boundaries in (4.24) and using (4.17) provides

$$A = \sqrt{\frac{1}{2a}}. \quad (4.25)$$

*The results that are obtained by considering an electron in a square well are similar to the ones which one receives when the wave mechanical properties of a **hydrogen atom** are calculated. As above, one considers an electron with charge $-e$ to be bound to its nucleus. The potential, V , in which the electron propagates is taken as the Coulombic potential $V = -e^2/(4\pi\epsilon_0 r)$. Since V is a function of the radius, r , the Schrödinger equation is more conveniently expressed in polar coordinates. Of main interest are, again, the conditions under which solutions to this Schrödinger equation exist. The treatment leads, similarly as above, to discrete energy levels:

$$E = -\frac{me^4}{2(4\pi\epsilon_0\hbar)^2} \frac{1}{n^2} = -13.6 \cdot \frac{1}{n^2} (\text{eV}). \quad (4.18a)$$

The main difference compared to the square well model is, however, that the energy is now proportional to $-1/n^2$ (and not to n^2 as in (4.18)). This results

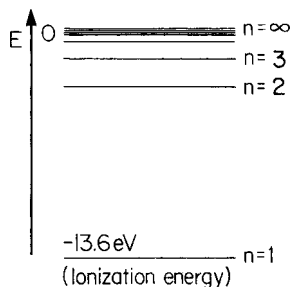


Figure 4.5. Energy levels of atomic hydrogen. E is the binding energy.

in a “crowding” of energy levels at higher energies. The energy at the lowest level is called the ionization energy, which has to be supplied to remove an electron from its nucleus. Energy diagrams, as in Fig. 4.5, are common in spectroscopy. The origin of the energy scale is arbitrarily set at $n = \infty$ and the ionization energies are counted negative. Since we are mainly concerned with the solid state, the detailed calculation of the hydrogen atom is not treated here.

*So far, we have considered the electron to be confined to a one-dimensional well. A similar calculation for a three-dimensional potential well (“**electron in a box**”) leads to an equation which is analogous to (4.18):

$$E_n = \frac{\hbar^2 \pi^2}{2ma^2} (n_x^2 + n_y^2 + n_z^2). \quad (4.26)$$

The smallest allowed energy in a three-dimensional potential well is occupied by an electron if $n_x = n_y = n_z = 1$. For the next higher energy there are three different possibilities for combining the n -values; namely, $(n_x, n_y, n_z) = (1, 1, 2), (1, 2, 1),$ or $(2, 1, 1)$. One calls the states which have the same energy but different quantum numbers “**degenerate**” states. The example just given describes a threefold degenerate energy state.

4.3. Finite Potential Barrier (Tunnel Effect)

Let us assume that a free electron, propagating in the positive x -direction, encounters a potential barrier whose potential energy, V_0 , (“height” of the barrier) is larger than the total energy, E , of the electron, but is still finite (Fig. 4.6). For this case we have to write two Schrödinger equations, which take into account the two different areas. In region I ($x < 0$) the electron is assumed to be free, and we can write

$$(I) \quad \frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2} E\psi = 0. \quad (4.27)$$

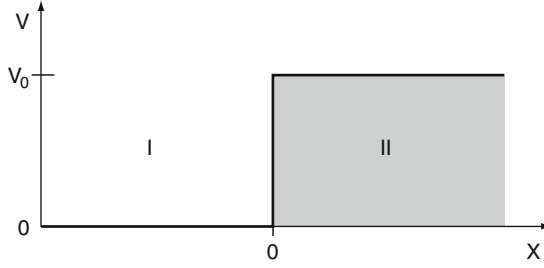


Figure 4.6. Finite potential barrier.

Inside the potential barrier ($x > 0$) the Schrödinger equation reads

$$(II) \quad \frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}(E - V_0)\psi = 0. \quad (4.28)$$

The solutions to these equations are as before (see Appendix 1):

$$(I) \quad \psi_I = Ae^{i\alpha x} + Be^{-i\alpha x}, \quad (4.29)$$

where

$$\alpha = \sqrt{\frac{2mE}{\hbar^2}}, \quad (4.30)$$

and

$$(II) \quad \psi_{II} = Ce^{i\beta x} + De^{-i\beta x}, \quad (4.31)$$

with

$$\beta = \sqrt{\frac{2m}{\hbar^2}(E - V_0)}. \quad (4.32)$$

A word of caution has to be inserted here. We stipulated above that V_0 is larger than E . As a consequence of this $(E - V_0)$ is negative and β becomes imaginary. To prevent this, we define a new parameter:

$$\gamma = i\beta. \quad (4.33)$$

This yields, for (4.32),

$$\gamma = \sqrt{\frac{2m}{\hbar^2}(V_0 - E)}. \quad (4.34)$$

The parameter γ is now prevented under the stated conditions from becoming imaginary. Rearranging (4.33) to obtain

$$\beta = \frac{\gamma}{i} \quad (4.35)$$

and inserting (4.35) into (4.31) yields

$$\psi_{II} = Ce^{\gamma x} + De^{-\gamma x}. \quad (4.36)$$

Next, one of the constants C or D needs to be determined by means of a boundary condition:

For $x \rightarrow \infty$ it follows from (4.36) that

$$\psi_{II} = C \cdot \infty + D \cdot 0. \quad (4.37)$$

The consequence of (4.37) could be that ψ_{II} and therefore $\psi_{II}\psi_{II}^*$ are infinity.

Since the probability $\psi\psi^*$ can never be larger than one (certainty), $\psi_{II} \rightarrow \infty$ is no solution. To avoid this, C has to go to zero:

$$C \rightarrow 0. \quad (4.38)$$

Then, (4.36) reduces to

$$\psi_{II} = De^{-\gamma x}, \quad (4.39)$$

which reveals that the ψ -function decreases in Region II exponentially, as shown in Fig. 4.7 (solid line). The decrease is stronger the larger γ is chosen, i.e., for a large potential barrier, V_0 .

The electron wave $\Psi(x, t)$ is then given, using (A.27) and (4.39), by

$$\Psi = De^{-\gamma x} \cdot e^{i(\omega t - kx)}$$

(damped wave) as shown by the dashed curve in Fig. 4.7. In other words, (4.39) provides the envelope (or decreasing amplitude) for the electron wave that propagates in the finite potential barrier. If the potential barrier is only moderately high and relatively narrow, the electron wave may continue on the opposite side of the barrier. This behavior is analogous to that for a light wave, which likewise penetrates to a certain degree into a material and whose amplitude also decreases exponentially, as we shall see in the optics part of this book, specifically in Fig. 10.4. The penetration of a potential barrier by an electron wave is called “**tunneling**” and has important applications in solid state physics (tunnel diode, tunnel electron microscope, field ion microscope). Tunneling is a quantum mechanical effect. In classical physics, the electron (particle) would be described to be entirely reflected back from the barrier (at $x = 0$) if its kinetic energy is smaller than V_0 .

*For the complete solution of the behavior of an electron wave that penetrates a finite potential barrier (Fig. 4.6), some additional boundary conditions need to be taken into consideration:

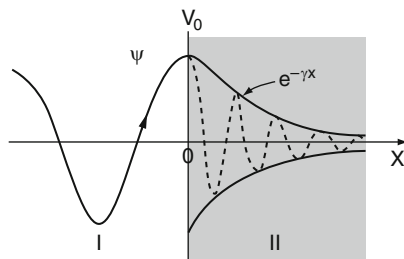


Figure 4.7. ψ -function (solid line) and electron wave (dashed line) meeting a finite potential barrier.

- (1) The functions ψ_I and ψ_{II} are continuous at $x = 0$. As a consequence, $\psi_I \equiv \psi_{II}$ at $x = 0$. This yields, with (4.29), (4.36), and (4.38),

$$Ae^{i\alpha x} + Be^{-i\alpha x} = De^{-\gamma x}.$$

With $x = 0$, we obtain

$$A + B = D. \quad (4.40)$$

- (2) The slopes of the wave functions in Regions I and II are continuous at $x = 0$, i.e., $(d\psi_I/dx) \equiv (d\psi_{II}/dx)$. This yields

$$Ai\alpha e^{i\alpha x} - Bi\alpha e^{-i\alpha x} = -\gamma De^{-\gamma x}. \quad (4.41)$$

With $x = 0$, one obtains

$$Ai\alpha - Bi\alpha = -\gamma D. \quad (4.42)$$

Inserting (4.40) into (4.42) yields

$$A = \frac{D}{2} \left(1 + i \frac{\gamma}{\alpha} \right) \quad (4.43)$$

and

$$B = \frac{D}{2} \left(1 - i \frac{\gamma}{\alpha} \right). \quad (4.43a)$$

From this, the ψ -functions can be expressed in terms of a constant D . Figure 4.8 illustrates the modification of Fig. 4.4(a) when tunneling is taken into consideration. A penetration of the ψ -function into the potential barriers is depicted.

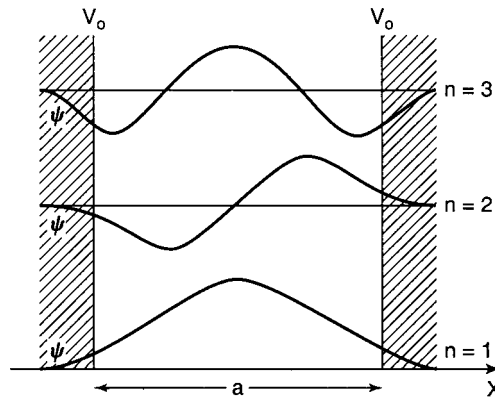


Figure 4.8. Square well with finite potential barriers. (The zero points on the vertical axis have been shifted for clarity.)

4.4. Electron in a Periodic Field of a Crystal (The Solid State)

In the preceding sections we became acquainted with some special cases, namely, the completely free electron and the electron which is confined to a potential well. The goal of this section is to study the behavior of an electron in a crystal. We will see eventually that the extreme cases which we treated previously can be derived from this general case.

Our first task is to find a potential distribution that is suitable for a solid. From high resolution transmission electron microscopy and from X-ray diffraction investigations, it is known that the atoms in a crystal are arranged periodically. Thus, for the treatment of our problem a periodic repetition of the potential well of Fig. 4.2, i.e., a periodic arrangement of potential wells and potential barriers, is probably close to reality and is also best suited for a calculation. Such a periodic potential is shown in Fig. 4.9 for the one-dimensional case.⁴

The potential distribution shows potential wells of length a , which we call Region I. These wells are separated by potential barriers of height V_0 and width b (Region II), where V_0 is assumed to be larger than the energy E of the electron.

This model is certainly a coarse simplification of the actual potential distribution in a crystal. It does not take into consideration that the inner electrons are more strongly bound to the core, i.e., that the potential function of a point charge varies as $1/r$. It also does not consider that the individual potentials from each lattice site overlap. A potential distribution which takes these features into consideration is shown in Fig. 4.10. It is immediately evident, however, that the latter model is less suitable for a simple calculation than the one which is shown in Fig. 4.9. Thus, we utilize the model shown in Fig. 4.9.

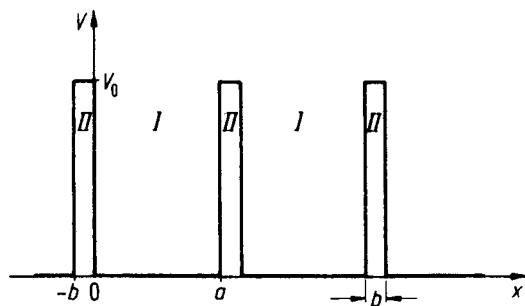


Figure 4.9. One-dimensional periodic potential distribution (simplified) (**Kronig-Penney model**).

⁴R. De. L. Kronig and W.G. Penney, *Proc. Roy. Soc. London*, **130**, 499 (1931).

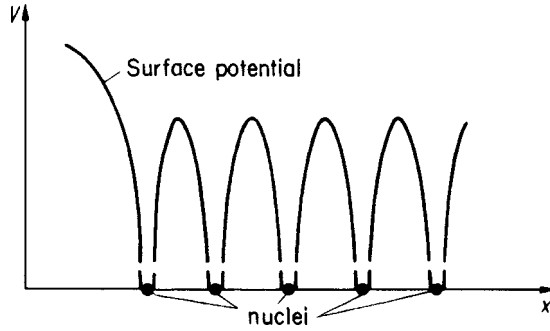


Figure 4.10. One-dimensional periodic potential distribution for a crystal (**muffin tin potential**).

We now write the Schrödinger equation for Regions I and II:

$$(I) \quad \frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2} E\psi = 0, \quad (4.44)$$

$$(II) \quad \frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2} (E - V_0)\psi = 0. \quad (4.45)$$

For abbreviation we write, as before,

$$\alpha^2 = \frac{2m}{\hbar^2} E, \quad (4.46)$$

and

$$\gamma^2 = \frac{2m}{\hbar^2} (V_0 - E). \quad (4.47)$$

(γ^2 is chosen in a way to keep it from becoming imaginary, see Section 4.3.) Equations (4.44) and (4.45) need to be solved simultaneously, a task which can be achieved only with considerable mathematical effort. Bloch⁵ showed that the solution of this type of equation has the following form:

$$\psi(x) = u(x) \cdot e^{ikx} \quad (4.48)$$

(Bloch function), where $u(x)$ is a periodic function which possesses the periodicity of the lattice in the x -direction. Therefore, $u(x)$ is no longer a constant (amplitude A) as in (4.2), but changes periodically with increasing x (modulated amplitude). Of course, $u(x)$ is different for various directions in the crystal lattice.

⁵F. Bloch, *Z. Phys.* **52**, 555 (1928); **59**, 208 (1930).

The reader who is basically interested in the results, and their implications for the electronic structure of crystals, may skip the mathematical treatment given below and refer directly to (4.67).

Differentiating the Bloch function (4.48) twice with respect to x provides

$$\frac{d^2\psi}{dx^2} = \left(\frac{d^2u}{dx^2} + \frac{du}{dx} 2ik - k^2 u \right) e^{ikx}. \quad (4.49)$$

We insert (4.49) into (4.44) and (4.45) and take into account the abbreviations (4.46) and (4.47):

$$(I) \quad \frac{d^2u}{dx^2} + 2ik \frac{du}{dx} - (k^2 - \alpha^2)u = 0, \quad (4.50)$$

$$(II) \quad \frac{d^2u}{dx^2} + 2ik \frac{du}{dx} - (k^2 + \gamma^2)u = 0. \quad (4.51)$$

Equations (4.50) and (4.51) have the form of an equation of a damped vibration. The solution⁶ to (4.50) and (4.51) is

$$(I) \quad u = e^{-ikx} (Ae^{i\alpha x} + Be^{-i\alpha x}), \quad (4.55)$$

$$(II) \quad u = e^{-ikx} (Ce^{-\gamma x} + De^{\gamma x}). \quad (4.56)$$

We have four constants A , B , C , and D which we need to dispose of by means of four boundary conditions: The functions ψ and $d\psi/dx$ pass over continuously from Region I into Region II at the point $x = 0$. Equation I = Equation II for $x = 0$ yields

$$A + B = C + D. \quad (4.57)$$

(du/dx) for I = (du/dx) for II at $x = 0$ provides

$$A(i\alpha - ik) + B(-i\alpha - ik) = C(-\gamma - ik) + D(\gamma - ik). \quad (4.58)$$

Further, ψ , and therefore u , is continuous at the distance $(a + b)$. This means that Equation I at $x = 0$ must be equal to Equation II at $x = a + b$,

⁶Differential equation of a damped vibration for spatial periodicity (see Appendix 1)

$$\frac{d^2u}{dx^2} + D \frac{du}{dx} + Cu = 0. \quad (4.52)$$

Solution:

$$u = e^{-(D/2)x} (Ae^{i\delta x} + Be^{-i\delta x}), \quad (4.53)$$

where

$$\delta = \sqrt{C - \frac{D^2}{4}}. \quad (4.54)$$

or, more simply, Equation I at $x = a$ is equal to Equation II at $x = -b$ (see Fig. 4.9). This yields

$$Ae^{(i\alpha-ik)a} + Be^{(-i\alpha-ik)a} = Ce^{(ik+\gamma)b} + De^{(ik-\gamma)b}. \quad (4.59)$$

Finally, (du/dx) is periodic in $a + b$:

$$Ai(\alpha - k)e^{ia(\alpha-k)} - Bi(\alpha + k)e^{-ia(\alpha+k)} = -C(\gamma + ik)e^{(ik+\gamma)b} + D(\gamma - ik)e^{(ik-\gamma)b}. \quad (4.60)$$

The constants A , B , C , and D can be determined by means of these four equations which, when inserted in (4.55) and (4.56), provide values for u . This also means that solutions for the function ψ can be given by using (4.48). However, as in the preceding sections, the knowledge of the ψ function is not of primary interest. We are searching instead for a condition which tells us where solutions to the Schrödinger equations (4.44) and (4.45) exist. We recall that these limiting conditions were leading to the energy levels in Section 4.2. We proceed here in the same manner. We use the four equations (4.57)–(4.60) and eliminate the four constants A – D . (This can be done by simple algebraic manipulation or by forming the determinant out of the coefficients A – D and equating this determinant to zero). The lengthy calculation provides, using some Euler equations,⁷

$$\frac{\gamma^2 - \alpha^2}{2\alpha\gamma} \sinh(\gamma b) \cdot \sin(\alpha a) + \cosh(\gamma b) \cos(\alpha a) = \cos k(a + b). \quad (4.61)$$

For simplification of the discussion of this equation we make the following stipulation. The potential barriers in Fig. 4.9 will be of the kind such that b is very small and V_0 is very large. It is further assumed that the product $V_0 b$, i.e., the area of this potential barrier, remains finite. In other words, if V_0 grows, b diminishes accordingly. The product $V_0 b$ is called the potential barrier strength.

If V_0 is very large, then E in (4.47) can be considered to be small compared to V_0 and can therefore be neglected so that

$$\gamma = \sqrt{\frac{2m}{\hbar^2}} \sqrt{V_0}. \quad (4.62)$$

Multiplication of (4.62) by b yields

$$\gamma b = \sqrt{\frac{2m}{\hbar^2}} \sqrt{(V_0 b)b}. \quad (4.63)$$

Since $V_0 b$ has to remain finite (see above) and $b \rightarrow 0$ it follows that γb becomes very small. For a small γb we obtain (see tables of the hyperbolic functions)

⁷See Appendix 2.

$$\cosh(\gamma b) \approx 1 \quad \text{and} \quad \sinh(\gamma b) \approx \gamma b. \quad (4.64)$$

Finally, one can neglect α^2 compared to γ^2 and b compared to a (see (4.46), (4.47), and Fig. 4.9) so that (4.61) reads as follows:

$$\frac{m}{\alpha \hbar^2} V_0 b \sin \alpha a + \cos \alpha a = \cos ka. \quad (4.65)$$

With the abbreviation

$$P = \frac{maV_0b}{\hbar^2} \quad (4.66)$$

we finally get from (4.65)

$$P \frac{\sin \alpha a}{\alpha a} + \cos \alpha a = \cos ka. \quad (4.67)$$

This is the desired relation which provides the allowed solutions to the Schrödinger equations (4.44) and (4.45). We notice that the boundary conditions lead to an equation with trigonometric function similarly as in Section 4.2. Therefore, only certain values of α are possible. This in turn means, because of (4.46), that only certain values for the energy E are defined. One can assess the situation best if one plots the function $P(\sin \alpha a / \alpha a) + \cos \alpha a$ versus αa , which is done in Fig. 4.11 for $P = (3/2)\pi$. It is of particular significance that the right-hand side of (4.67) allows only certain values of this function because $\cos ka$ is only defined between $+1$ and -1 (except for imaginary k -values). This is shown in Fig. 4.11, in which the allowed values of the function $P(\sin \alpha a / \alpha a) + \cos \alpha a$ are marked by heavy lines on the αa -axis.

We arrive herewith at the following very important result: Because αa is a function of the energy, the above-mentioned limitation means that an electron that moves in a periodically varying potential field can only occupy certain allowed energy zones. Energies outside of these allowed zones or

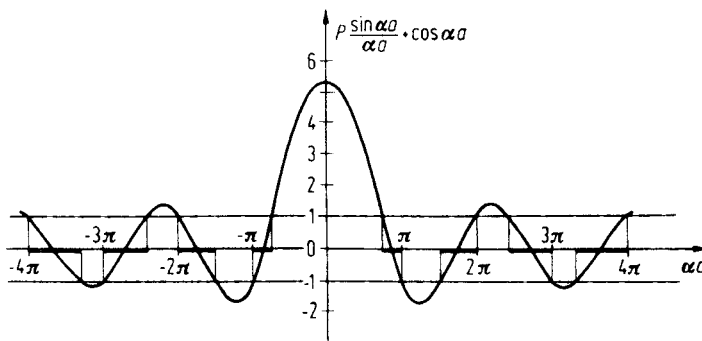


Figure 4.11. Function $P(\sin \alpha a / \alpha a) + \cos \alpha a$ versus αa . P was arbitrarily set to be $(3/2)\pi$.

“bands” are prohibited. One sees from Fig. 4.11 that with increasing values of αa (i.e., with increasing energy), the disallowed (or forbidden) bands become narrower. The size of the allowed and forbidden energy bands varies with the variation of P . Below, four special cases will be discussed.

- (a) If the “**potential barrier strength**” $V_0 b$ (see Fig. 4.9) is large, then, according to (4.66), P is also large and the curve in Fig. 4.11 proceeds more steeply. The allowed bands are narrow.
- (b) If the potential barrier strength, and therefore P , is small, the allowed bands become wider (see Fig. 4.12).
- (c) If the potential barrier strength becomes smaller and smaller and finally disappears completely, P goes toward zero, and one obtains from (4.67)

$$\cos \alpha a = \cos ka \quad (4.68)$$

or $\alpha = k$. From this it follows, with (4.46), that

$$E = \frac{\hbar^2 k^2}{2m}.$$

This is the well-known equation (4.8) for free electrons which we derived in Section 4.1.

- (d) If the potential barrier strength is very large, P approaches infinity. However, because the left-hand side of (4.67) has to stay within the limits ± 1 , i.e., it has to remain finite, it follows that

$$\frac{\sin \alpha a}{\alpha a} \rightarrow 0,$$

i.e., $\sin \alpha a \rightarrow 0$. This is only possible if $\alpha a = n\pi$ or

$$\alpha^2 = \frac{n^2 \pi^2}{a^2} \quad \text{for } n = 1, 2, 3, \dots \quad (4.69)$$

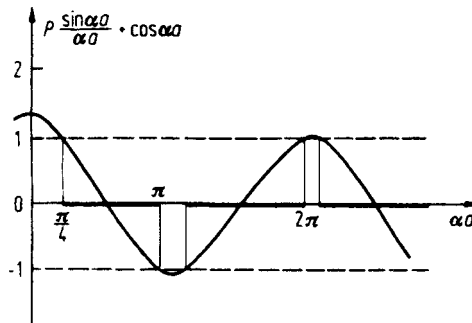


Figure 4.12. Function $P(\sin \alpha a / \alpha a) + \cos \alpha a$ with $P = \pi/10$.

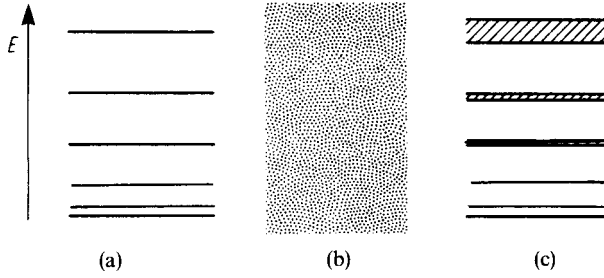


Figure 4.13. Allowed energy levels for (a) bound electrons, (b) free electrons, and (c) electrons in a solid.

Combining (4.46) with (4.69) yields

$$E = \frac{\pi^2 \hbar^2}{2ma^2} \cdot n^2,$$

which is the result of Section 4.2, equation (4.18).

We summarize (Fig. 4.13): If the electrons are strongly bound, i.e., if the potential barrier is very large, one obtains sharp energy levels (electron in the potential field of *one* ion). If the electron is not bound, one obtains a continuous energy region (free electrons). If the electron moves in a periodic potential field, one receives energy bands (solid).

The widening of the energy levels into energy bands and the transition into a quasi-continuous energy region is shown in Fig. 4.14. This widening occurs because the atoms increasingly interact as their separation distance decreases. The arrows *a*, *b*, and *c* refer to the three sketches of Fig. 4.13.

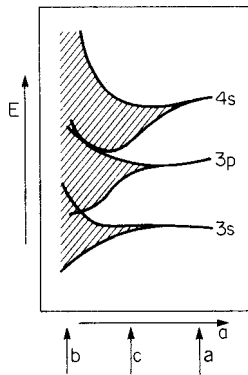


Figure 4.14. Widening of the sharp energy levels into bands and finally into a quasi-continuous energy region with decreasing interatomic distance, a , for a metal (after calculations of Slater). The quantum numbers are explained in Appendix 3.

Problems

- Describe the energy for:
 - a free electron;
 - a strongly bound electron; and
 - an electron in a periodic potential.
 Why do we get these different band schemes?

2. *Computer problem.* Plot $\psi\psi^*$ for an electron in a potential well. Vary n from 1 to ~ 100 . What conclusions can be drawn from these graphs? (*Hint:* If for large values for n you see strange periodic structures, then you need to choose more data points!)

- State the two Schrödinger equations for electrons in a periodic potential field (Kronig–Penney model). Use for their solutions, instead of the Bloch function, the trial solution

$$\psi(x) = Ae^{ikx}.$$

Discuss the result. (*Hint:* For free electrons $V_0 = 0$.)

*4. When treating the Kronig–Penney model, we arrived at four equations for the constants A , B , C , and D . Confirm (4.61).

- The differential equation for an undamped vibration is

$$a \frac{d^2 u}{dx^2} + bu = 0, \quad (1)$$

whose solution is

$$u = Ae^{ikx} + Be^{-ikx}, \quad (2)$$

where

$$k = \sqrt{b/a}. \quad (3)$$

Prove that (2) is indeed a solution of (1).

- Calculate the “ionization energy” for atomic hydrogen.
- Derive (4.18a) in a semiclassical way by assuming that the centripetal force of an electron, mv^2/r , is counterbalanced by the **Coulombic attraction force**, $-e^2/4\pi\epsilon_0 r^2$, between the nucleus and the orbiting electron. Use Bohr’s postulate which states that the angular momentum $L = mvr$ (v = linear electron velocity and r = radius of the orbiting electron) is a multiple integer of Planck’s constant (i.e., $n \cdot \hbar$). (*Hint:* The kinetic energy of the electron is $E = \frac{1}{2}mv^2$.)
- Computer problem.* Plot equation (4.67) and vary values for P .
- Computer problem.* Plot equation (4.39) for various values for D and γ .
- The width of the potential well (Fig. 4.2) of an electron can be assumed to be about $2 \text{ \AA}$. Calculate the energy of an electron (in Joules and in eV) from this information for various values of n . Give the zero-point energy.

CHAPTER 5

Energy Bands in Crystals

5.1. One-Dimensional Zone Schemes

We are now in a position to make additional important statements which contribute considerably to the understanding of the properties of crystals. For this we plot the energy versus the momentum of the electrons, or, because of (4.8), versus the wave vector, $\mathbf{k}$. As before, we first discuss the one-dimensional case.

The relation between E and k_x is particularly simple in the case of free electrons, as can be seen from (4.8),

$$k_x = \text{const. } E^{1/2}. \quad (5.1)$$

The plot of E versus k_x is a parabola (Fig. 5.1).

We return now to (4.68), which we obtained from (4.67) for $P = 0$ (free electrons). Because the cosine function is periodic in 2π , (4.68) should be written in the more general form

$$\cos \alpha a = \cos k_x a \equiv \cos(k_x a + n2\pi), \quad (5.2)$$

where $n = 0, \pm 1, \pm 2, \dots$. This gives

$$\alpha a = k_x a + n2\pi. \quad (5.3)$$

Combining (4.8),

$$\alpha = \sqrt{\frac{2m}{\hbar^2}} E^{1/2},$$

with (5.3) yields

$$k_x + n \frac{2\pi}{a} = \sqrt{\frac{2m}{\hbar^2}} E^{1/2}. \quad (5.4)$$

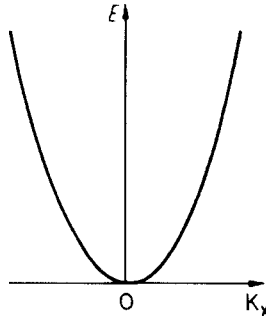


Figure 5.1. Electron energy E versus the wave vector k_x for free electrons.

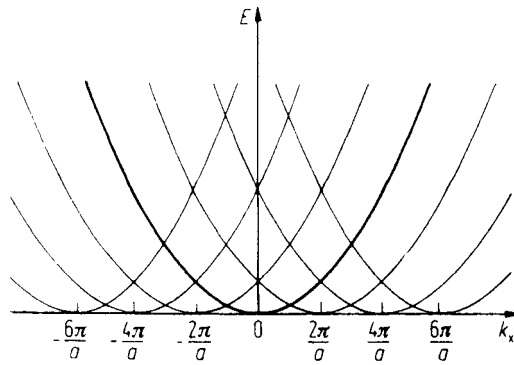


Figure 5.2. Periodic repetition of Fig. 5.1 at the points $k_x = n \cdot 2\pi/a$. The figure depicts a family of free electron parabolas having a periodicity of $\pm 2\pi/a$.

We see from (5.4) that in the general case the parabola, shown in Fig. 5.1, is repeated periodically in intervals of $n \cdot 2\pi/a$ (Fig. 5.2). The energy is thus a periodic function of k_x with the periodicity $2\pi/a$.

We noted, when discussing Fig. 4.11, that if an electron propagates in a periodic potential we always observe discontinuities of the energies when $\cos k_x a$ has a maximum or a minimum, i.e., when $\cos k_x a = \pm 1$. This is only the case when

$$k_x a = n\pi, \quad n = \pm 1, \pm 2, \pm 3, \dots, \quad (5.5)$$

or

$$k_x = n \cdot \frac{\pi}{a}. \quad (5.6)$$

At these singularities, a deviation from the parabolic E versus k_x curve occurs, and the branches of the individual parabolas merge into the neighboring ones.⁸ This is shown in Fig. 5.3.

⁸If two energy functions with equal symmetry cross, the quantum mechanical “noncrossing rule” requires that the eigenfunctions be split, so that they do not cross.

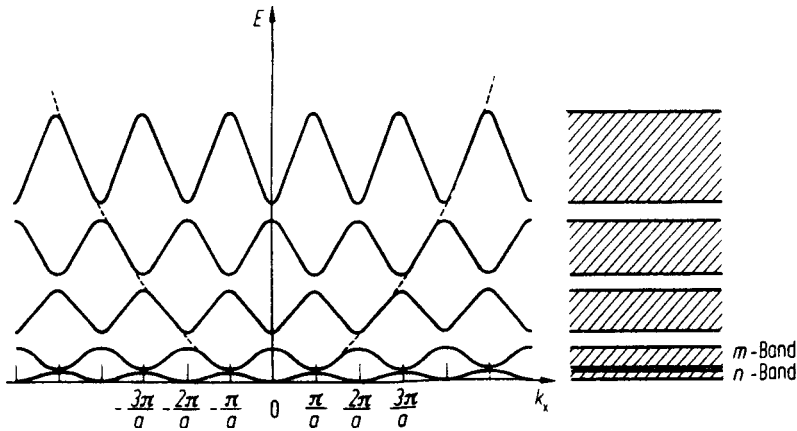
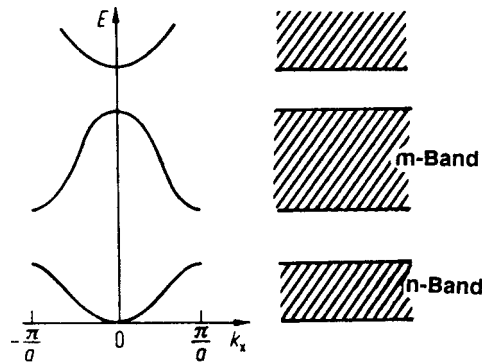


Figure 5.3. Periodic zone scheme.

Figure 5.4. Reduced zone scheme. (This is a section of Fig. 5.3 between $-\pi/a$ and $+\pi/a$.)

The aforementioned consideration leads to a very important result. The electrons in a crystal behave, for most k_x values, like free electrons, except when k_x approaches the value $n \cdot \pi/a$.

Besides this “**periodic zone scheme**” (Fig. 5.3), two further zone schemes are common. In the future we will use mostly the “**reduced zone scheme**” (Fig. 5.4), which is a section of Fig. 5.3 between the limits $\pm\pi/a$. In the “**extended zone scheme**” (Fig. 5.5), the deviations from the free electron parabola at the critical points $k_x = n \cdot \pi/a$ are particularly easy to identify.

Occasionally, it is useful to plot *free* electrons in a reduced zone scheme. In doing so, one considers the width of the forbidden bands to be reduced until the energy gap between the individual branches disappears completely. This leads to the “**free electron bands**” which are shown in Fig. 5.6 for a special case. The well-known band character disappears for free electrons, however, and one obtains a continuous energy region as explained in Section 4.1. As before, the shape of the individual branches in Fig. 5.6 is

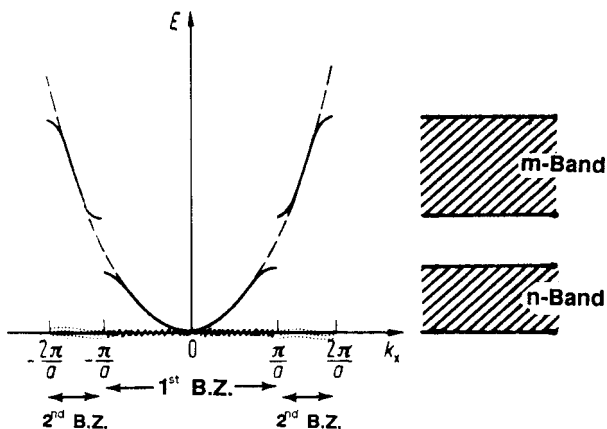


Figure 5.5. Extended zone scheme. The first and second Brillouin zones (BZ) are shown, see Section 5.2.

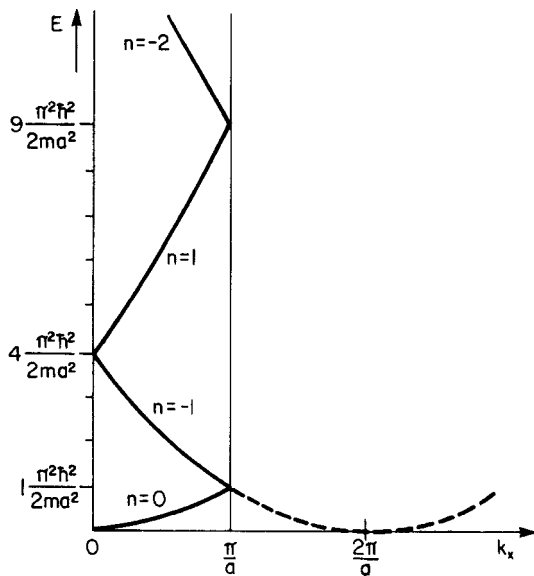


Figure 5.6. "Free electron bands" plotted in the reduced zone scheme (cubic primitive crystal structure). Compare this figure with the central portion of Fig. 5.2, that is, with the region from zero to π/a . Note the sameness of the individual bands.

due to the $2\pi/a$ periodicity, as a comparison with Fig. 5.2 shows. From (5.4), it follows that

$$E = \frac{\hbar^2}{2m} \left(k_x + n \frac{2\pi}{a} \right)^2, \quad n = 0, \pm 1, \pm 2, \dots \quad (5.7)$$

By inserting different n -values in (5.7), one can calculate the shape of the branches of the free electron bands. A few examples might illustrate this:

$$\mathbf{n} = \mathbf{0} \quad \text{yields } E = \frac{\hbar^2}{2m} k_x^2 (\text{parabola with 0 as origin});$$

$$\mathbf{n} = -\mathbf{1} \quad \text{yields } E = \frac{\hbar^2}{2m} \left(k_x - \frac{2\pi}{a} \right)^2 \left(\text{parabola with } \frac{2\pi}{a} \text{ as origin} \right);$$

$$\text{specifically, for } k_x = 0 \text{ follows } E = 4 \frac{\pi^2 \hbar^2}{2ma^2};$$

$$\text{and for } k_x = \frac{\pi}{a} \text{ follows } E = 1 \frac{\pi^2 \hbar^2}{2ma^2}.$$

The calculated data are depicted in Fig. 5.6. (The calculation of the remaining branches (bands) is left to the reader, see Problem 5.)

One important question has remained essentially unanswered: What do these E versus $|\mathbf{k}|$ curves really mean? Simply stated, they relate the energy of an electron to its $\mathbf{k}$ -vector, i.e., with its momentum. They provide in principle quite similar information as, for example, a distance versus time diagram for a moving car, or a “stress-strain diagram” in mechanical metallurgy, or a “phase diagram” in materials science. All these diagrams relate in graphic form one parameter with another variable in order to provide an easier interpretation of data. We shall eventually learn to appreciate complete band diagrams in later chapters, from which we will draw important conclusions about the electronic properties of materials.

In Figs. 5.3, 5.4, and 5.5 the individual allowed energy regions and the disallowed energy regions, called **band gaps**, are clearly seen. We call the allowed bands, for the time being, the n -band, or the m -band, and so forth. In later sections and particularly in semiconductor physics (see Chapter 8) we will call one of these bands the **valence band** (because it contains the valence electrons) and the next higher one the **conduction band**.

An additional item needs to be mentioned: It is quite common to use the word “band” for both the allowed energy regions, such as the n -band or the m -band, as well as for the individual branches within a band as seen, for example, in Fig. 5.6. As a rule this does not cause any confusion.

Finally, we need to stress one more point: The wave vector $\mathbf{k}$ is inversely proportional to the wavelength of the electrons (see equation (4.9)). Thus, $\mathbf{k}$ has the unit of a reciprocal length and is therefore defined in “**reciprocal space**.” The reader might recall from a course in crystallography that each crystal structure has two lattices associated with it, one of them being the crystal (or real) lattice and the other the reciprocal lattice. We will show in Section 5.5 how these two lattices are related. The following may suffice for the moment: each lattice plane in real space can be represented by a vector which is normal to this plane and whose length is made proportional to the

reciprocal of the interplanar distance. The tips of all such vectors from sets of parallel lattice planes form the points in a reciprocal lattice. An X-ray diffraction pattern is a map of such a reciprocal lattice.

5.2. One- and Two-Dimensional Brillouin Zones

Let us again inspect Fig. 5.5. We noticed there that the energy versus k_x curve, between the boundaries $-\pi/a$ and $+\pi/a$, corresponds to the first electron band, which we arbitrarily labeled as *n-band*. This region in k -space between $-\pi/a$ and $+\pi/a$ is called the first **Brillouin zone (BZ)**. Accordingly, the area between π/a and $2\pi/a$, and also between $-\pi/a$ and $-2\pi/a$, which corresponds to the *m-band*, is called the second Brillouin zone. In other words, the lowest band shown in Fig. 5.5 corresponds to the first Brillouin zone, the next higher band corresponds to the second Brillouin zone, and so on. Now, we learned above that the individual branches in an extended zone scheme (Fig. 5.5) are $2\pi/a$ periodic, i.e., they can be shifted by $2\pi/a$ to the left or to the right. We make use of this concept and shift the branch of the second Brillouin zone on the positive side of the $E - (k_x)$ diagram in Fig. 5.5 by $2\pi/a$ to the left, and likewise the left band of the second Brillouin zone by $2\pi/a$ to the right. A reduced zone scheme as shown in Fig. 5.4 is the result. Actually, we projected the second Brillouin zone into the first Brillouin zone. The same can be done with the third Brillouin zone, etc. This has very important implications: we do not need to plot E versus k -curves for *all* Brillouin zones; the relevant information is, because of the $2\pi/a$ periodicity, already contained in the first Brillouin zone, i.e., in a reduced zone scheme.

We now consider the behavior of an electron in the potential of a two-dimensional lattice. The electron movement in two dimensions can be described as before by the wave vector $\mathbf{k}$ that has the components k_x and k_y , which are parallel to the x - and y -axes in reciprocal space. Points in the $k_x - k_y$ coordinate system form a two-dimensional reciprocal lattice (see Fig. 5.7). One obtains, in the two-dimensional case, a two-dimensional field of allowed energy regions which corresponds to the allowed energy bands, i.e., one obtains two-dimensional Brillouin zones.

We shall illustrate the construction of the Brillouin zones for a two-dimensional reciprocal lattice (Fig. 5.7). For the first zone one constructs the perpendicular bisectors on the shortest lattice vectors, G_1 . The area that is enclosed by these four “Bragg planes” is the first Brillouin zone. For the following zones the bisectors of the next shortest lattice vectors are constructed. It is essential that for the zones of higher order the extended limiting lines of the zones of lower order are used as additional limiting lines. The first four Brillouin zones are shown in Fig. 5.8. Note that all the zones have the same area. The first four shortest lattice vectors G_1 through G_4 are drawn in Fig. 5.7.

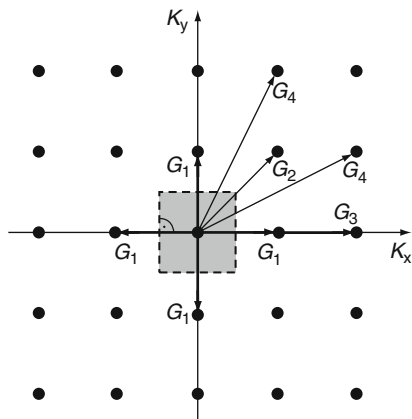


Figure 5.7. Four shortest lattice vectors in a $k_x - k_y$ coordinate system and the first Brillouin zone in a two-dimensional reciprocal lattice. (Cubic primitive crystal structure.)

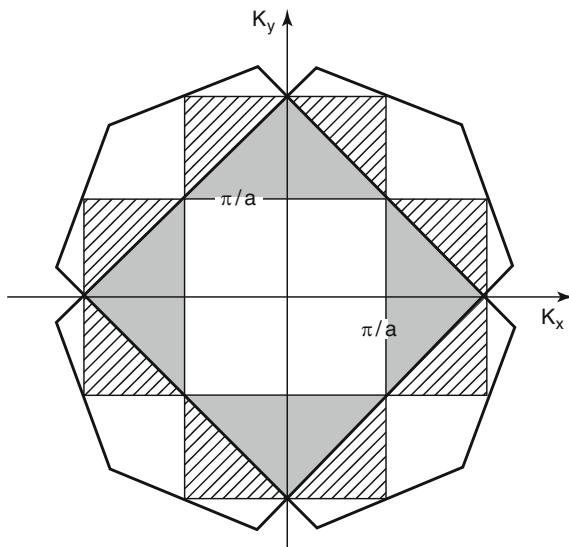


Figure 5.8. The first four Brillouin zones of a two-dimensional, cubic primitive reciprocal lattice.

The significance of the Brillouin zones will become evident in later sections, when the energy bands of solids are discussed. A few words of explanation will be given here, nevertheless. The Brillouin zones are useful if one wants to calculate the behavior of an electron which may travel in a specific direction in reciprocal space. For example, if in a two-dimensional lattice an electron travels at 45° to the k_x -axis, then the boundary of the Brillouin zone is reached, according to Fig. 5.8, for

$$k_{\text{crit}} = \frac{\pi}{a} \sqrt{2}. \quad (5.8)$$

This yields with (4.8) a maximal attainable energy of

$$E_{\text{max}} = \frac{\hbar^2}{2m} k_{\text{crit}}^2 = \frac{\pi^2 \hbar^2}{a^2 m}. \quad (5.8a)$$

On the other hand, the boundary of a Brillouin zone is reached at

$$k_{\text{crit}} = \frac{\pi}{a} \quad (5.9)$$

when an electron moves parallel to the k_x - or k_y -axes. The largest energy that electrons can assume in this second case is only

$$E_{\text{max}} = \frac{1}{2} \left(\frac{\pi^2 \hbar^2}{a^2 m} \right). \quad (5.9a)$$

Once the maximal energy has been reached, the electron waves form standing waves (or equivalently, the electrons are reflected back into the Brillouin zone).

The consequence of (5.8) and (5.9) is an overlapping of energy bands which can be seen when the bands are drawn in different directions in k -space (Fig. 5.9). We will learn later that these considerations can be utilized to determine the difference between metals, semiconductors, and insulators.

*The occurrence of critical energies at which a reflection of the electron wave takes place can also be illustrated in a completely different way. This will be done briefly here because of its immediate intuitive power. We consider an electron wave that propagates in a lattice at an angle θ to a set of parallel lattice planes (Fig. 5.10). The corresponding rays are diffracted on the lattice atoms. At a certain angle of incidence, constructive interference between rays 1' and 2' occurs. It has been shown by Bragg that each ray

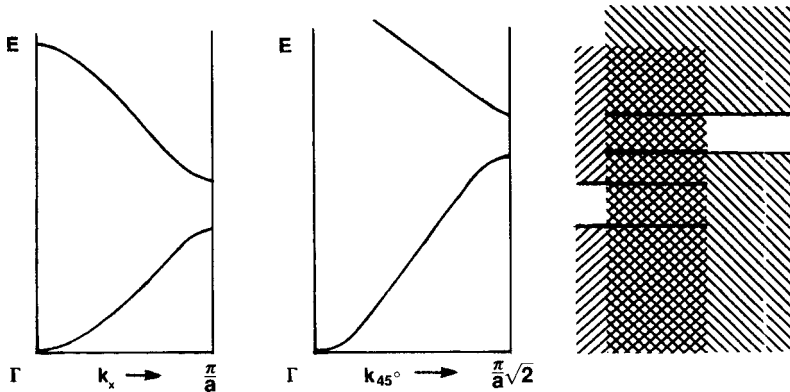


Figure 5.9. Overlapping of allowed energy bands.

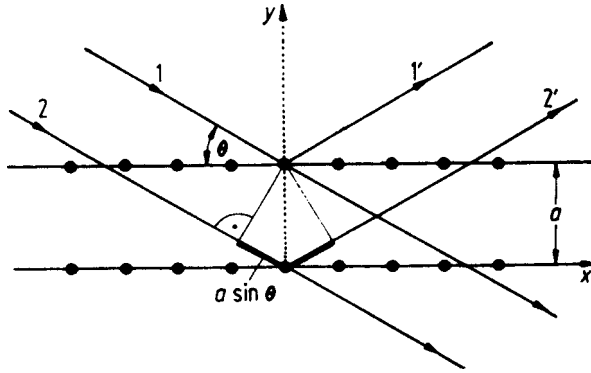


Figure 5.10. Bragg reflection of an electron wave in a lattice. The angle of incidence is θ .

which is diffracted in this way can be considered as being reflected by a mirror parallel to the lattice planes. In other words, at a critical angle the “reflected” rays will be enhanced considerably. This is always the case when the path difference $2a \sin \theta$ is an integer multiple of the electron wavelength λ , i.e., when

$$2a \sin \theta = n\lambda, \quad n = 1, 2, 3, \dots \quad (5.10)$$

(Bragg relation). With (4.9) one obtains, from (5.10),

$$2a \sin \theta = n \frac{2\pi}{k}$$

and therefore

$$k_{\text{crit}} = n \frac{\pi}{a \sin \theta}. \quad (5.11)$$

For perpendicular incidence ($\theta = 90^\circ$) equation (5.11) becomes (5.9). On the other hand, if $\theta = 45^\circ$, one obtains (5.8).

Equation (5.11) leads to the result that for increasing electron energies a critical $\mathbf{k}$ -value is finally reached for which “reflection” of the electron wave at the lattice planes occurs. At this critical $\mathbf{k}$ -value the transmission of an electron beam through the lattice is prevented. Then, the incident and the Bragg-reflected electron wave form a standing wave.

*5.3. Three-Dimensional Brillouin Zones

In the previous section, the physical significance of the Brillouin zones was discussed. It was shown that at the boundaries of these zones the electron waves are Bragg-reflected by the crystal. The wave vector $|\mathbf{k}| = 2\pi/\lambda$ was

seen to have the unit of a reciprocal length and is therefore defined in the reciprocal lattice. We will now attempt to construct three-dimensional Brillouin zones for two important crystal structures, namely, the face-centered cubic (fcc) and the body-centered cubic (bcc) crystals. Since the Brillouin zones for these structures have some important features in common with the so-called Wigner–Seitz cells, it is appropriate to discuss, at first, the Wigner–Seitz cells and also certain features of the reciprocal lattice before we return to the Brillouin zones at the end of Section 5.5.

*5.4. Wigner–Seitz Cells

Crystals have symmetrical properties. Therefore, a crystal can be described as an accumulation of “**unit cells**.” In general, the smaller such a unit cell, i.e., the fewer atoms it contains, the simpler its description. The smallest possible cell is called a “**primitive unit cell**.” Frequently, however, a larger, nonprimitive unit cell is used, which might have the advantage that the symmetry can be better recognized. Body-centered cubic and face-centered cubic are characteristic representatives of such “**conventional**” unit cells.⁹

The Wigner–Seitz cell is a special type of primitive unit cell that shows the cubic symmetry of the cubic cells. For its construction, one bisects the vectors from a given atom to its nearest neighbors and places a plane perpendicular to these vectors at the bisecting points. This is shown in Fig. 5.11 for the bcc lattice.

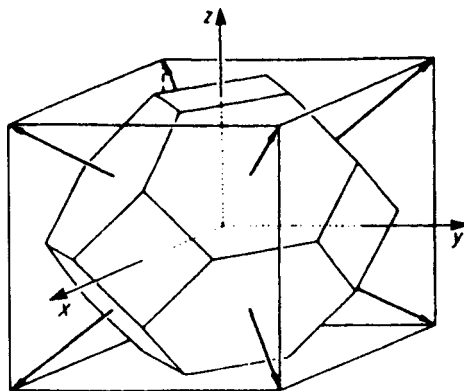


Figure 5.11. Wigner–Seitz cell for the body-centered cubic (bcc) structure.

⁹A lattice is a regular periodic arrangement of points in space; it is, consequently, a mathematical abstraction. All crystal structures can be traced to one of the 14 types of **Bravais lattices** (see textbooks on crystallography).

In the fcc lattice, the atoms are arranged on the corners and faces of a cube, which is equivalent to the center points of the edges and the center of the cell (Fig. 5.12). The Wigner–Seitz cell for this structure is shown in Fig. 5.13.

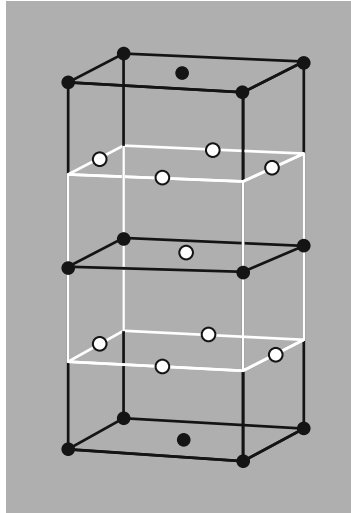


Figure 5.12. Conventional unit cell of the fcc structure. In the cell which is marked black, the atoms are situated on the corners and faces of the cubes. In the white cell, the atoms are at the centers of the edges and the center of the cell.

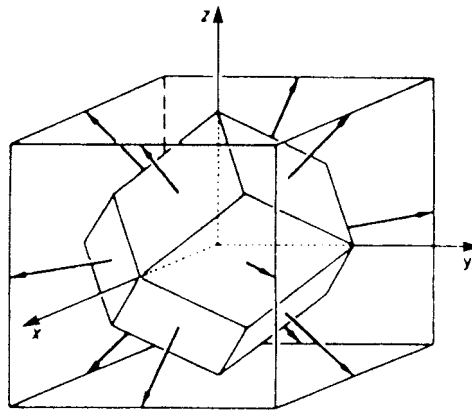


Figure 5.13. Wigner–Seitz cell for the fcc structure. It is constructed from the white cell which is marked in Fig. 5.12.

*5.5. Translation Vectors and the Reciprocal Lattice

In Fig. 5.14(a) the fundamental vectors $\mathbf{t}_1$, $\mathbf{t}_2$, $\mathbf{t}_3$ are inserted in a unit cell of a cubic primitive lattice. By combination of these “**primitive vectors**” a **translation vector**,

$$\mathbf{R} = n_1 \mathbf{t}_1 + n_2 \mathbf{t}_2 + n_3 \mathbf{t}_3, \quad (5.12)$$

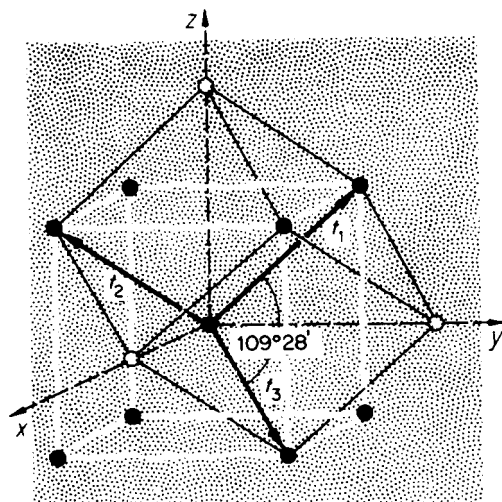
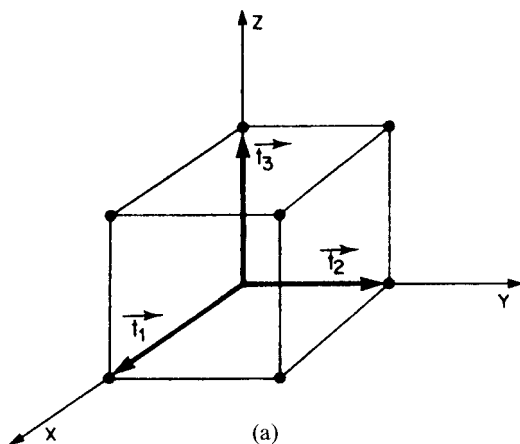


Figure 5.14. (a) Fundamental lattice vectors $\mathbf{t}_1$, $\mathbf{t}_2$, $\mathbf{t}_3$ in a cubic primitive lattice. (b) Fundamental lattice vectors in a conventional (white) and primitive, noncubic unit cell (black) of a bcc lattice. The axes of the primitive (noncubic) unit cell form angles of $109^\circ 28'$.

can be defined. Using this translation vector it is possible to reach, from a given lattice point, any other equivalent lattice point. For this, the factors n_1, n_2, n_3 have to be integers. In Fig. 5.14(b) the fundamental vectors $\mathbf{t}_1, \mathbf{t}_2, \mathbf{t}_3$ are shown in a conventional unit cell of a bcc lattice.

Similarly, as above, we now introduce for the reciprocal lattice three vectors, $\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3$, and a translation vector

$$\mathbf{G} = 2\pi(h_1\mathbf{b}_1 + h_2\mathbf{b}_2 + h_3\mathbf{b}_3), \quad (5.13)$$

where h_1, h_2 , and h_3 are, again, integers. (The factor 2π is introduced for convenience. In X-ray crystallography, this factor is omitted.)

The real and reciprocal lattices are related by a definition which states that the scalar product of the vectors $\mathbf{t}_1$ and $\mathbf{b}_1$ should be unity, whereas the scalar products of $\mathbf{b}_1$ and $\mathbf{t}_2$ or $\mathbf{b}_1$ and $\mathbf{t}_3$ are zero:

$$\mathbf{b}_1 \cdot \mathbf{t}_1 = 1, \quad (5.14)$$

$$\mathbf{b}_1 \cdot \mathbf{t}_2 = 0, \quad (5.15)$$

$$\mathbf{b}_1 \cdot \mathbf{t}_3 = 0. \quad (5.16)$$

Equivalent equations are defined for $\mathbf{b}_2$ and $\mathbf{b}_3$. These nine equations can be combined by using the **Kronecker-Delta symbol**,

$$\mathbf{b}_n \cdot \mathbf{t}_m = \delta_{nm}, \quad (5.17)$$

where $\delta_{nm} = 1$ for $n = m$ and $\delta_{nm} = 0$ for $n \neq m$. Equation (5.17) is from now on our *definition* for the three vectors $\mathbf{b}_m$, which are reciprocal to the vectors $\mathbf{t}_m$. From (5.15) and (5.16) it follows¹⁰ that $\mathbf{b}_1$ is perpendicular to $\mathbf{t}_2$ and to $\mathbf{t}_3$, which means that $\mathbf{t}_2$ and $\mathbf{t}_3$ form a plane perpendicular to the vector $\mathbf{b}_1$ (Fig. 5.15). We therefore write¹¹

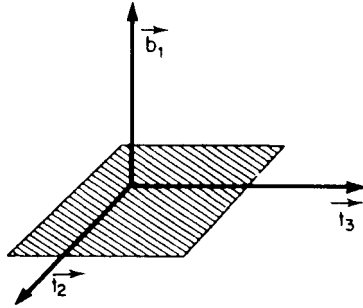


Figure 5.15. Plane formed by $\mathbf{t}_2$ and $\mathbf{t}_3$ with perpendicular vector $\mathbf{b}_1$.

¹⁰The *scalar product* of two vectors $\mathbf{a}$ and $\mathbf{b}$ is $\mathbf{a} \cdot \mathbf{b} = ab \cos(\mathbf{ab})$. If $\mathbf{i}, \mathbf{j}$, and $\mathbf{l}$ are mutually perpendicular unit vectors, then we can write $\mathbf{i} \cdot \mathbf{j} = \mathbf{j} \cdot \mathbf{l} = \mathbf{l} \cdot \mathbf{i} = 0$ and $\mathbf{i} \cdot \mathbf{i} = \mathbf{j} \cdot \mathbf{j} = \mathbf{l} \cdot \mathbf{l} = 1$.

¹¹The *vector product* of two vectors $\mathbf{a}$ and $\mathbf{b}$ is a vector which stands perpendicular to the plane formed by $\mathbf{a}$ and $\mathbf{b}$. It is $\mathbf{i} \times \mathbf{i} = \mathbf{j} \times \mathbf{j} = \mathbf{l} \times \mathbf{l} = 0$ and $\mathbf{i} \times \mathbf{j} = \mathbf{l}$ and $\mathbf{j} \times \mathbf{i} = -\mathbf{l}$.

$$\mathbf{b}_1 = \text{const. } \mathbf{t}_2 \times \mathbf{t}_3. \quad (5.18)$$

To evaluate the constant, we form the scalar product of $\mathbf{t}_1$ and $\mathbf{b}_1$ (5.18) and make use of (5.14):

$$\mathbf{b}_1 \cdot \mathbf{t}_1 = \text{const. } \mathbf{t}_1 \cdot \mathbf{t}_2 \times \mathbf{t}_3 = 1. \quad (5.19)$$

This yields

$$\text{const.} = \frac{1}{\mathbf{t}_1 \cdot \mathbf{t}_2 \times \mathbf{t}_3}. \quad (5.20)$$

Combining (5.18) with (5.20) gives

$$\mathbf{b}_1 = \frac{\mathbf{t}_2 \times \mathbf{t}_3}{\mathbf{t}_1 \cdot \mathbf{t}_2 \times \mathbf{t}_3}. \quad (5.21)$$

Equivalent equations can be obtained for $\mathbf{b}_2$ and $\mathbf{b}_3$:

$$\mathbf{b}_2 = \frac{\mathbf{t}_3 \times \mathbf{t}_1}{\mathbf{t}_1 \cdot \mathbf{t}_2 \times \mathbf{t}_3}, \quad (5.22)$$

$$\mathbf{b}_3 = \frac{\mathbf{t}_1 \times \mathbf{t}_2}{\mathbf{t}_1 \cdot \mathbf{t}_2 \times \mathbf{t}_3}. \quad (5.23)$$

Equations (5.21)–(5.23) are the transformation equations which express the fundamental vectors $\mathbf{b}_1$, $\mathbf{b}_2$, and $\mathbf{b}_3$ of the reciprocal lattice in terms of real lattice vectors.

As an example of how these transformations are performed, we calculate now the reciprocal lattice of a bcc crystal. The real crystal may have the lattice constant “a.” We express the lattice vectors $\mathbf{t}_1$, $\mathbf{t}_2$, $\mathbf{t}_3$ in terms of the unit vectors, $\mathbf{i}$, $\mathbf{j}$, $\mathbf{l}$ in the x , y , z coordinate system (see Fig. 5.14(b)):

$$\mathbf{t}_1 = \frac{a}{2}(-\mathbf{i} + \mathbf{j} + \mathbf{l}), \quad (5.24)$$

or, abbreviated,

$$\mathbf{t}_1 = \frac{a}{2}(\bar{1}11) \quad (5.25)$$

and

$$\mathbf{t}_2 = \frac{a}{2}(1\bar{1}1), \quad (5.26)$$

$$\mathbf{t}_3 = \frac{a}{2}(11\bar{1}). \quad (5.27)$$

To calculate $\mathbf{b}_1$, using (5.21), we form at first the vector product¹²

¹² $\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{l} \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix}.$

$$\begin{aligned} \mathbf{t}_2 \times \mathbf{t}_3 &= \frac{a^2}{4} \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{l} \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix} = \frac{a^2}{4} (\mathbf{i} + \mathbf{j} + \mathbf{l} + \mathbf{l} - \mathbf{i} + \mathbf{j}) = \frac{a^2}{4} (2\mathbf{j} + 2\mathbf{l}) \\ &= \frac{a^2}{2} (\mathbf{j} + \mathbf{l}) \end{aligned} \quad (5.28)$$

and the scalar¹³ product

$$\mathbf{t}_1 \cdot \mathbf{t}_2 \times \mathbf{t}_3 = \frac{a^3}{4} (-\mathbf{i} + \mathbf{j} + \mathbf{l}) \cdot (0 + \mathbf{j} + \mathbf{l}) = \frac{a^3}{4} (0 + 1 + 1) = \frac{a^3}{2}. \quad (5.29)$$

Combining (5.21) with (5.28) and (5.29) yields

$$\mathbf{b}_1 = \frac{\frac{a^2}{2} (\mathbf{j} + \mathbf{l})}{\frac{a^3}{2}} = \frac{1}{a} (\mathbf{j} + \mathbf{l}), \quad (5.30)$$

or

$$\mathbf{b}_1 = \frac{1}{a} (011). \quad (5.31)$$

Similar calculations yield

$$\mathbf{b}_2 = \frac{1}{a} (101), \quad (5.32)$$

$$\mathbf{b}_3 = \frac{1}{a} (110). \quad (5.33)$$

In Fig. 5.16, the vectors $\mathbf{b}_1$, $\mathbf{b}_2$, $\mathbf{b}_3$ are inserted into a cube of length $2/a$. We note immediately an important result. The end points of the reciprocal lattice vectors of a bcc crystal are at the center of the edges of a cube. This means that points of the reciprocal lattice of the bcc structure are identical to the lattice points in a real lattice of the fcc structure, see Fig. 5.12. Conversely, the reciprocal lattice points of the fcc structure and the real lattice points of the bcc structure are identical.

In Section 5.2, we constructed two-dimensional Brillouin zones by drawing perpendicular bisectors on the shortest lattice vectors. Similarly, a three-dimensional Brillouin zone can be obtained by bisecting all lattice vectors $\mathbf{b}$ and placing planes perpendicular on these points. As has been shown in Section 5.4, this construction is identical for a Wigner–Seitz cell. A comparison of the fundamental lattice vectors $\mathbf{b}$ and $\mathbf{t}$ gives the striking result that the Wigner–Seitz cell for an fcc crystal (Fig. 5.13) and the first Brillouin zone for a bcc crystal (Fig. 5.17) are identical in shape. The same is true for

¹³ $\mathbf{a} \cdot \mathbf{b} = a_x b_x + a_y b_y + a_z b_z$.

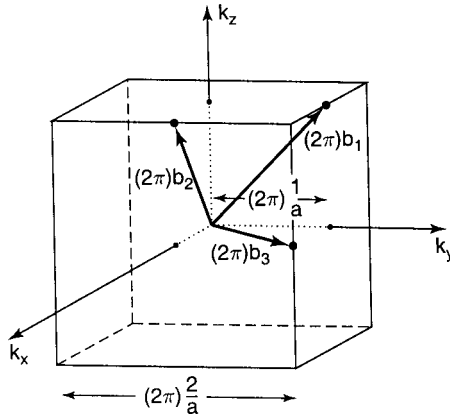


Figure 5.16. Lattice vectors in reciprocal space of a bcc crystal. The primitive vectors in the reciprocal lattice are (because of (5.13)) larger by a factor of 2π . The lattice constant of the cube then becomes $2\pi \cdot 2/a$.

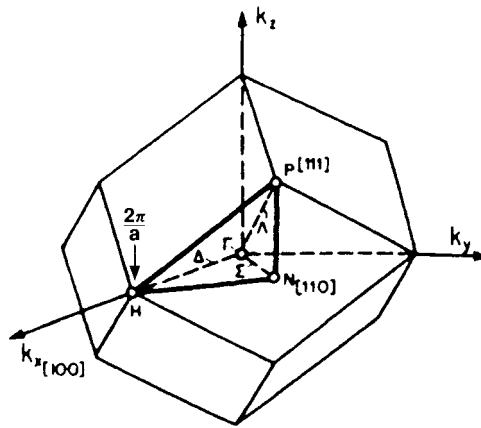


Figure 5.17. First Brillouin zone of the bcc crystal structure.

the Wigner–Seitz cell for bcc and the first Brillouin zone for fcc. Thus, a Brillouin zone can be defined as a Wigner–Seitz cell in the reciprocal lattice.

From (5.31) it can again be seen that the reciprocal lattice vector has the unit of a reciprocal length.

*5.6. Free Electron Bands

We mentioned in Section 5.1 that, because of the $E(\mathbf{k})$ periodicity, all information pertaining to the electronic properties of materials is contained in the first Brillouin zone. In other words, the energy $E_{\mathbf{k}'}$ for $\mathbf{k}'$ outside the

first zone is identical to the energy $E_{\mathbf{k}}$ within the first zone if a suitable translation vector $\mathbf{G}$ can be found so that a wave vector $\mathbf{k}'$ becomes

$$\mathbf{k}' = \mathbf{k} + \mathbf{G}. \quad (5.34)$$

We have already used this feature in Section 5.1, where we plotted one-dimensional energy bands in the form of a reduced zone scheme. We proceed now to three-dimensional zone pictures. We might correctly expect that the energy bands are not alike in different directions in $\mathbf{k}$ -space. This can be demonstrated by using the “free electron bands” which we introduced in Fig. 5.6. We explain the details using the bcc crystal structure as an example.

In three dimensions the equation analogous to (5.7) reads

$$E_{\mathbf{k}'} = \frac{\hbar^2}{2m} (\mathbf{k} + \mathbf{G})^2. \quad (5.35)$$

In Fig. 5.17 three important directions in $\mathbf{k}$ -space are inserted into the first Brillouin zone of a bcc lattice. They are the [100] direction from the origin (Γ) to point H , the [110] direction from Γ to N , and the [111] direction from Γ to P .¹⁴ These directions are commonly labeled by the symbols Δ , Σ , and Λ , respectively. Figure 5.18 depicts the bands, calculated by using (5.35), for these distinct directions in $\mathbf{k}$ -space. The sequence of the individual subgraphs is established by convention and can be followed using Fig. 5.17.

We now show how some of these bands are calculated for a simple case. We select the $\Gamma - H$ direction as an example. We vary the modulus of the vector $\mathbf{k}_{\Gamma H} \equiv \mathbf{k}_x$ between 0 and $2\pi/a$, the latter being the boundary of the Brillouin zone (see Fig. 5.16).¹⁵ For this direction, (5.35) becomes

$$E = \frac{\hbar^2}{2m} \left(\frac{2\pi}{a} x \mathbf{i} + \mathbf{G} \right)^2, \quad (5.36)$$

where x may take values between 0 and 1. To start with, let $\mathbf{G}$ be 0. Then (5.36) reads

$$E = \frac{\hbar^2}{2m} \left(\frac{2\pi}{a} \right)^2 (x\mathbf{i})^2 \equiv Cx^2 \quad (5.37)$$

¹⁴Directions in unit cells are identified by subtracting the coordinates of the tail from the coordinates of the tip of a distance vector. The set of numbers thus gained is inserted into square brackets; see textbooks on materials science.

¹⁵The attentive reader may have noticed that the boundary of the first Brillouin zone in the k_x direction for the bcc lattice is $2\pi/a$, and not π/a as for the cubic primitive unit cell (Fig. 5.6). This can be convincingly seen by comparing Figs. 5.13, 5.16, and 5.17.

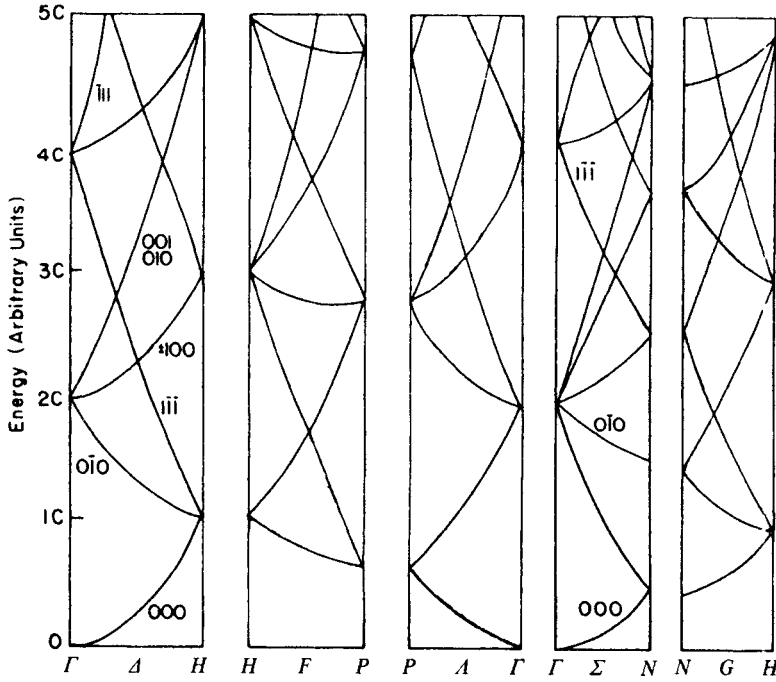


Figure 5.18. Energy bands of the free electrons for the bcc structure. The numbers given on the branches are the respective h_i values (see the calculation in the text). Compare to Fig. 5.6. $C = \hbar^2 2\pi^2 / ma^2$, see (5.38).

(see footnote 10), where

$$C = \frac{\hbar^2}{2m} \left(\frac{2\pi}{a} \right)^2 = \frac{2\hbar^2 \pi^2}{ma^2}. \quad (5.38)$$

This yields the well-known parabolic $E(\mathbf{k})$ -dependence. The curve which represents (5.37) is labeled (000) in Fig. 5.18, because h_1 , h_2 , and h_3 in (5.13) are all zero for $\mathbf{G} = 0$.

Now we let $h_1 = 0$, $h_2 = -1$, and $h_3 = 0$. Then we obtain, by using (5.13) and (5.32),

$$\mathbf{G} = -\frac{2\pi}{a}(\mathbf{i} + \mathbf{l}). \quad (5.39)$$

Combining (5.36) with (5.38) and (5.39) provides

$$\begin{aligned} E &= \frac{\hbar^2}{2m} \left[\frac{2\pi x}{a} \mathbf{i} - \frac{2\pi}{a}(\mathbf{i} + \mathbf{l}) \right]^2 = C[\mathbf{i}(x-1) - \mathbf{l}]^2 \\ &= C[(x-1)^2 + 1] = C(x^2 - 2x + 2) \end{aligned} \quad (5.40)$$

(see footnote 10), which yields for

$$x = 0 \rightarrow E = 2C$$

and for

$$x = 1 \rightarrow E = 1C.$$

We obtain the band labeled $(0\bar{1}0)$ in Fig. 5.18. Similarly, all bands in Fig. 5.18 can be calculated by variation of the h values and $\mathbf{k}$ -directions and by using (5.35).

The free electron bands are very useful for the following reason: by comparing them with the band structures of actual materials, an assessment is possible if and to what degree the electrons in that material can be considered to be free.

In Figs. 5.19 and 5.20 the first Brillouin zone and the free electron bands of the fcc structure are shown.

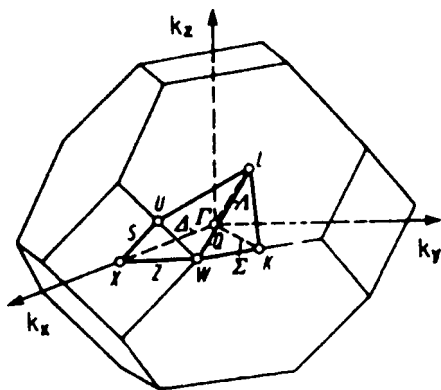


Figure 5.19. First Brillouin zone of the fcc structure.

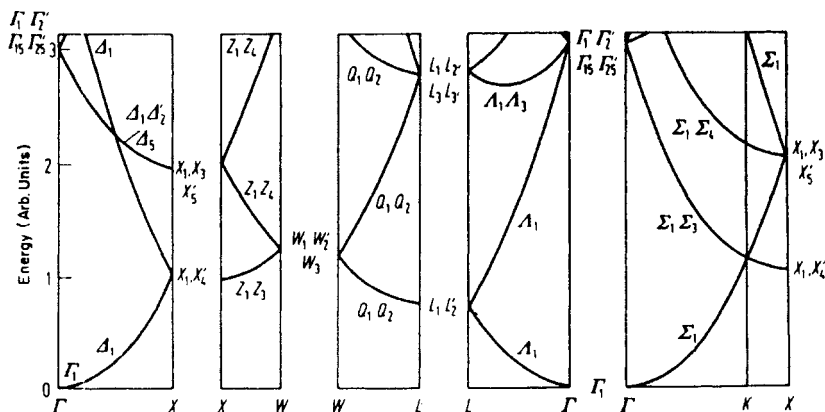


Figure 5.20. Free electron bands of the fcc structure. The letters on the bottom of the graphs correspond to letters in Fig. 5.19 and indicate specific symmetry points in k -space.

5.7. Band Structures for Some Metals and Semiconductors

Those readers who have skipped Sections 5.3 through 5.6 need to familiarize themselves with the (three-dimensional) first Brillouin zone for the face centered cubic (fcc) crystal structure (Fig. 5.19). The $[100]$, the $[110]$, and the $[111]$ directions in k -space are indicated by the letters $\Gamma - X$, $\Gamma - K$, and $\Gamma - L$, respectively. Other directions in k -space are likewise seen. These specific symmetry points and directions are selected by convention from a much larger number of possible directions. They sufficiently characterize the properties of materials, as we will see below.

We inspect now some calculated energy-band structures. They should resemble the one shown in Fig. 5.4. In the present case, however, they are depicted for more than one direction in k -space. Additionally, they are displayed in the *positive* k -direction only, similarly as in Fig. 5.6 or 5.20.

We start with the band diagram for **aluminum**, Fig. 5.21. We recognize immediately the characteristic parabola-shaped bands in the $k_x(\Gamma - X)$ direction as seen before in Fig. 5.4. Similar parabolic bands can be detected in the $\Gamma - K$ and the $\Gamma - L$ directions. The band diagram for aluminum looks quite similar to the free electron bands shown in Fig. 5.20. This suggests that the electrons in aluminum behave essentially free-electronlike (which is indeed the case).

We also detect in Fig. 5.21 some band gaps, for example, between the X'_4 and X_1 symmetry points, or between W_3 and W'_2 . Note, however, that the individual energy bands overlap in different directions in k -space, so that as a whole no band gap exists. (This is in marked difference to the band diagram of a semiconductor, as we shall see in a moment.) The lower,

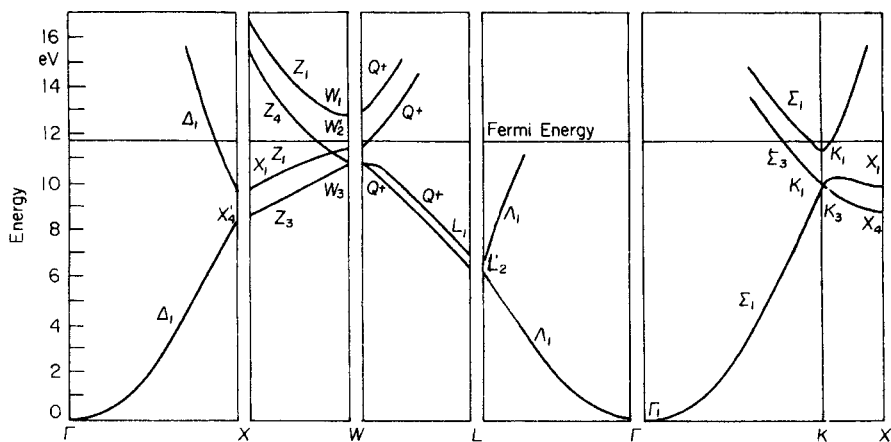


Figure 5.21. Energy bands for aluminum. Adapted from B. Segal, *Phys. Rev.* **124**, 1797 (1961). (The meaning of the Fermi energy will be explained in Section 6.1.)

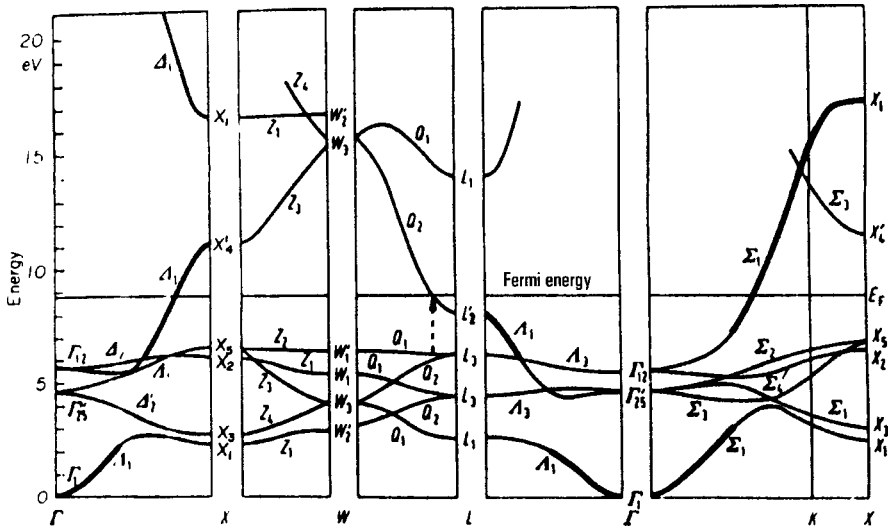


Figure 5.22. Band structure of copper (fcc). Adapted from B. Segal, *Phys. Rev.* **125**, 109 (1962). The calculation was made using the l -dependent potential. (For the definition of the Fermi energy, see Section 6.1.)

parabola-shaped bands are associated with the aluminum $3s$ electrons (see Appendix 3). These bands are therefore called “ $3s$ bands”. The origin of the energy scale is positioned for convenience in the lower end of this s -band.

Next, we discuss the band structure for **copper**, Fig. 5.22. We notice in the lower half of this diagram closely spaced and flat running bands. Calculations show that these can be attributed to the $3d$ -bands of copper (see Appendix 3). They superimpose the $4s$ -bands (which are heavily marked in Fig. 5.22). The band which starts at Γ is, at first, s -electronlike, and becomes d -electronlike while approaching point X . The first half of this band is continued at higher energies. It is likewise heavily marked. It can be seen, therefore, that the d -bands overlap the s -bands. Again, as for aluminum, no band gap exists if one takes all directions in $\mathbf{k}$ -space into consideration.

As a third example, the band structure of **silicon** is shown (Fig. 5.23). Of particular interest is the area between 0 and approximately 1 eV in which no energy bands are shown. This “energy gap,” which is responsible for the well-known semiconductor properties, will be the subject of detailed discussion in a later chapter. For semiconductors, the zero point of the energy scale is placed at the bottom of this energy gap, even though other conventions are possible and in use.

Finally, the band structure of **gallium arsenide** is shown in Fig. 5.24. The so-called III–V semiconductor compounds, such as GaAs, are of great technical importance for optoelectronic devices, as we will discuss in later sections. They have essentially the same crystal structure and the same total

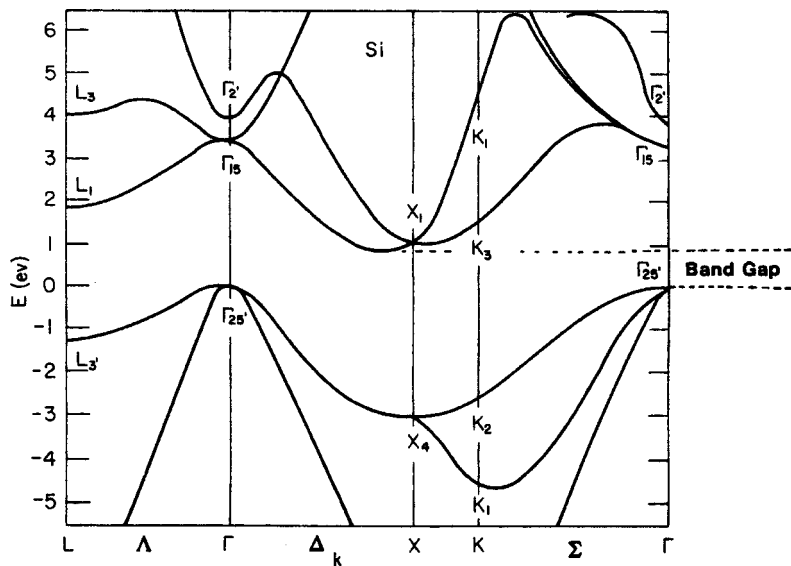


Figure 5.23. Calculated energy band structure of silicon (diamond-cubic crystal structure). Adapted from M.L. Cohen and T.K. Bergstresser, *Phys. Rev.* **14**, 789 (1966). See also J.R. Chelikowsky and M.L. Cohen, *Phys. Rev.* **B14**, 556 (1976).

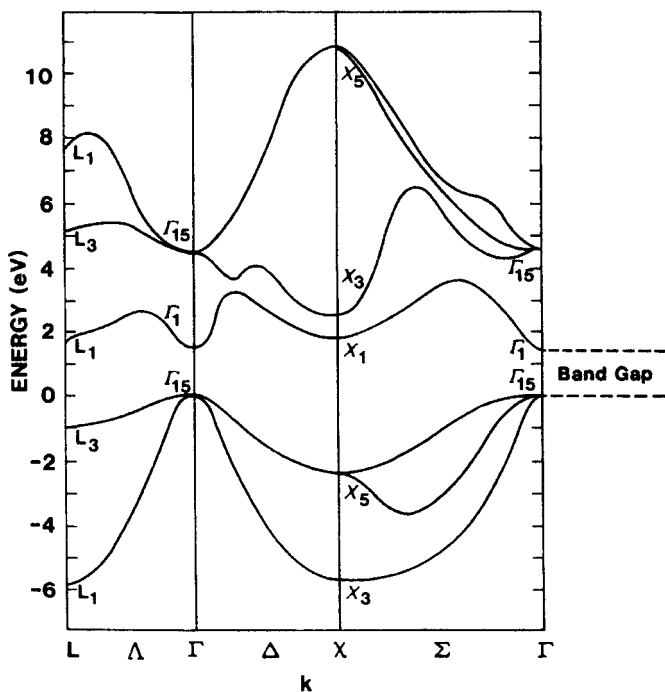


Figure 5.24. Calculated energy band structure of GaAs. Adapted from F. Herman and W.E. Spicer, *Phys. Rev.* **174**, 906 (1968).

number of valence electrons as the element silicon. Again, a band gap is clearly seen.

It should be mentioned, in closing, that the band structures of actual solids, as shown in Figs. 5.21–5.24, are the result of extensive, computer-aided calculations, and that various investigators using different starting potentials arrive at slightly different band structures. Experimental investigations, such as measurements of the frequency dependence of the optical properties, can help determine which of the various calculated band structures are closest to reality.

5.8. Curves and Planes of Equal Energy

We conclude this chapter by discussing another interesting aspect of the energy versus wave vector relationship.

In one-dimensional $\mathbf{k}$ -“space” there is only *one* (positive) $\mathbf{k}$ -value which is connected with a given energy (see Fig. 5.1). In the two-dimensional case, i.e., when we plot the electron energy over a $k_x - k_y$ plane, more than one $\mathbf{k}$ -value can be assigned to a given energy. This leads to curves of equal energy, as shown in Fig. 5.25. For a two-dimensional square lattice and for small electron energies, the curves of equal energy are circles. However, if the energy of the electrons is approaching the energy of the boundary of a Brillouin zone, then a deviation from the circular form is known to occur. This is shown in Fig. 5.26, where curves of equal energy for a two-dimensional square lattice are inserted into the first Brillouin zone. It is of particular interest that the energy which belongs to point K in Fig. 5.26

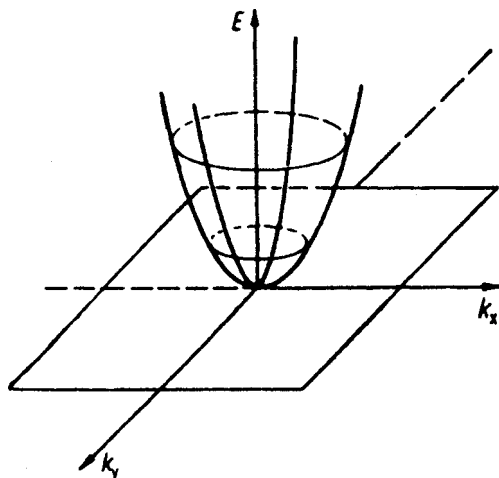


Figure 5.25. Electron energy E versus wave vector $\mathbf{k}$ (two-dimensional). This figure demonstrates various curves of equal energy for free electrons.

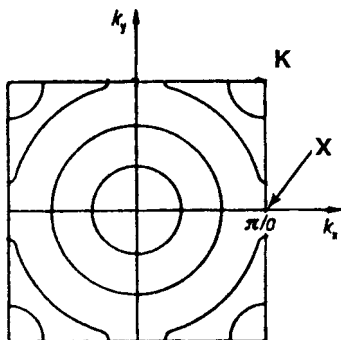


Figure 5.26. Curves of equal energy inserted into the first Brillouin zone for a two-dimensional square lattice.

is larger than the energy which belongs to point X (see (5.8a) and (5.9a)). Consequently, the curves of equal energy for the first Brillouin zone may extend into the second zone. This leads to an overlapping of energy bands as schematically shown in Fig. 5.9, and in the band structures of Figs. 5.21–5.24. For copper and aluminum the band overlapping leads to quasi-continuous allowed energies (in different directions of $\mathbf{k}$ -space). For semiconductors the band overlapping is not complete, which results in the already-mentioned energy gap (Figs. 5.23 and 5.24).

In three-dimensional $\mathbf{k}$ -space one obtains *surfaces* of equal energy. For the free electron case and for a cubic lattice they are spheres. For a nonparabolic $E(\mathbf{k})$ behavior these surfaces become more involved. This is demonstrated in Fig. 5.27 for a special case.

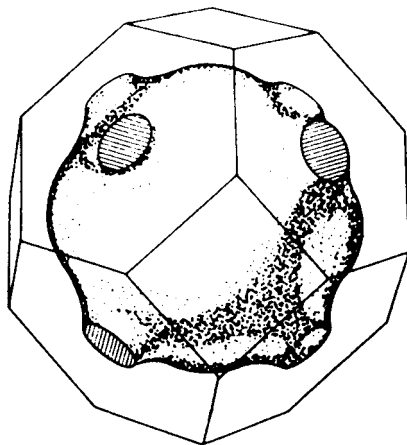


Figure 5.27. A particular surface of equal energy (Fermi surface, see Section 6.1) and the first Brillouin zone for copper. Adapted from A.B. Pippard, *Phil. Trans. Roy. Soc. London*, A 250, 325 (1957).

Problems

1. What is the energy difference between the points L'_2 and L_1 (upper) in the band diagram for copper?
2. How large is the “gap energy” for silicon? (*Hint*: Consult the band diagram for silicon.)
3. Calculate how much the kinetic energy of a free electron at the corner of the first Brillouin zone of a simple cubic lattice (three dimensions!) is larger than that of an electron at the midpoint of the face.
4. Construct the first four Brillouin zones for a simple cubic lattice in two dimensions.
5. Calculate the shape of the free electron bands for the cubic primitive crystal structure for $n = 1$ and $n = -2$ (see Fig. 5.6).
6. Calculate the free energy bands for a bcc structure in the $\mathbf{k}_x$ -direction having the following values for $h_1/h_2/h_3$: (a) $\overline{1}\overline{1}\overline{1}$; (b) 001; and (c) 010. Plot the bands in $\mathbf{k}$ -space. Compare with Fig. 5.18.
7. Calculate the main lattice vectors in reciprocal space of an fcc crystal.
8. Calculate the bands for the bcc structure in the 110 $[\Gamma - N]$ direction for: (a) (000); (b) $(0\overline{1}0)$; and (c) $\overline{1}\overline{1}\overline{1}$.
9. If $\mathbf{b}_1 \cdot \mathbf{t}_1 = 1$ is given (see equation (5.14)), does this mean that $\mathbf{b}_1$ is parallel to $\mathbf{t}_1$?

